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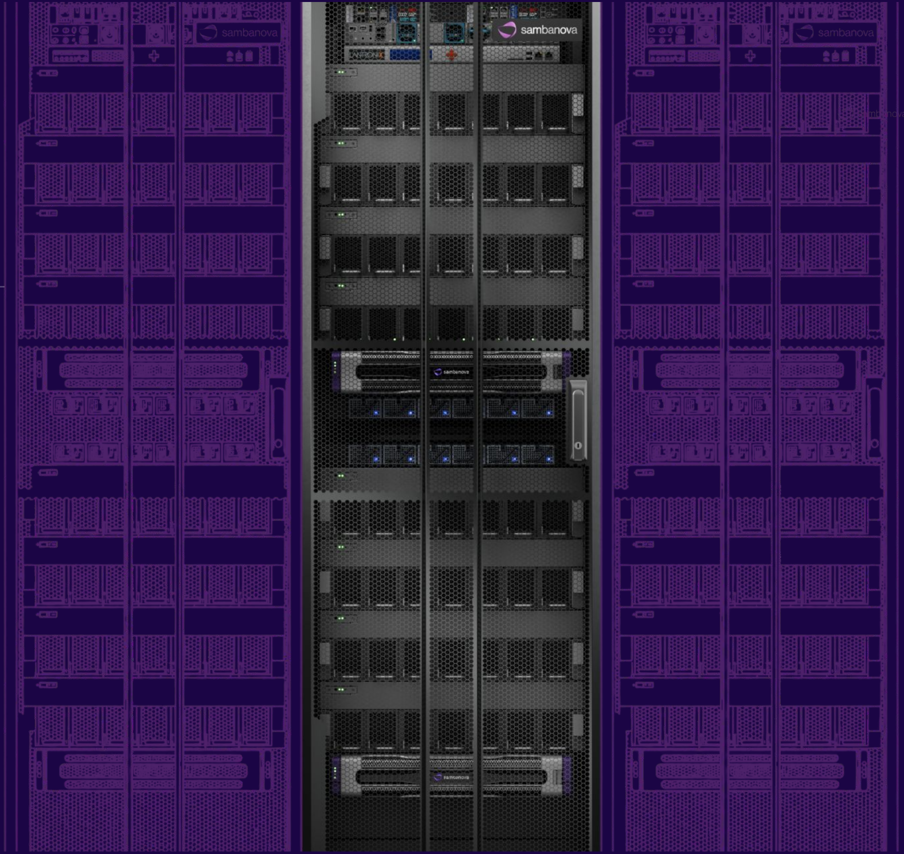
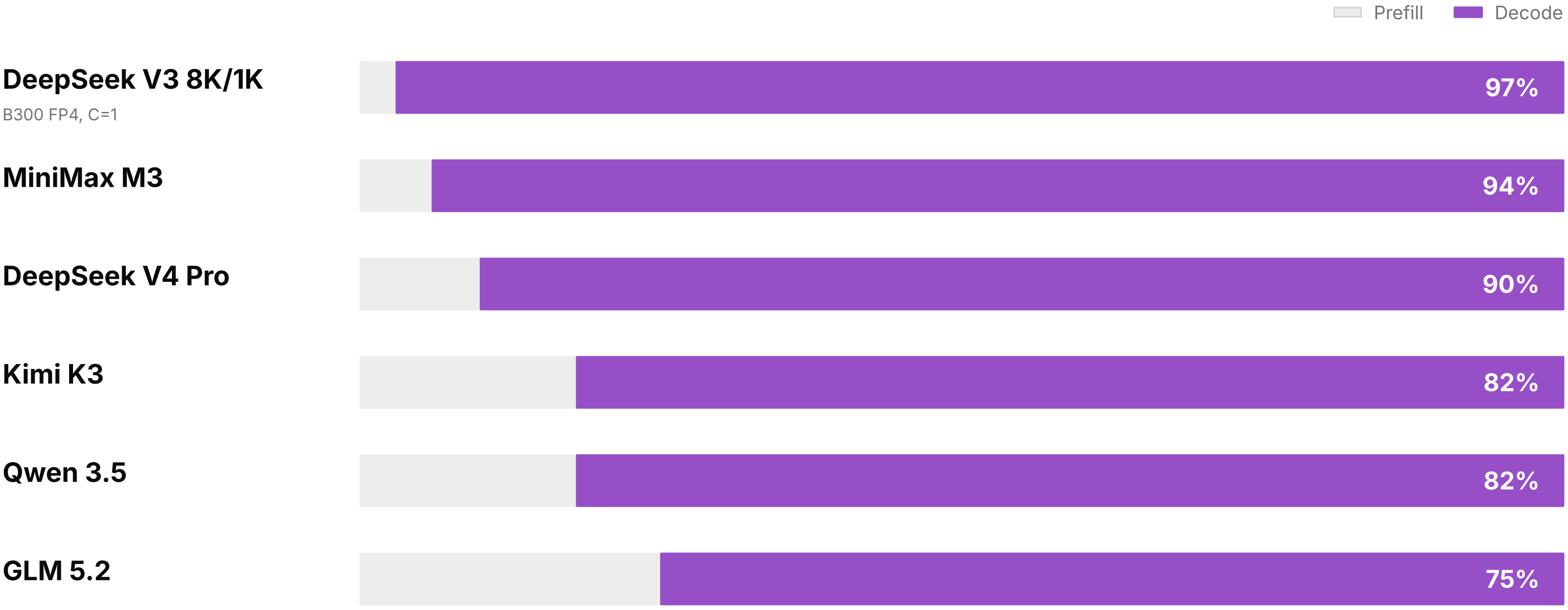
Dataflow at Scale: the SN50 RDU

Raghu Prabhakar
Chief Architect



Agentic inference is decode-dominated

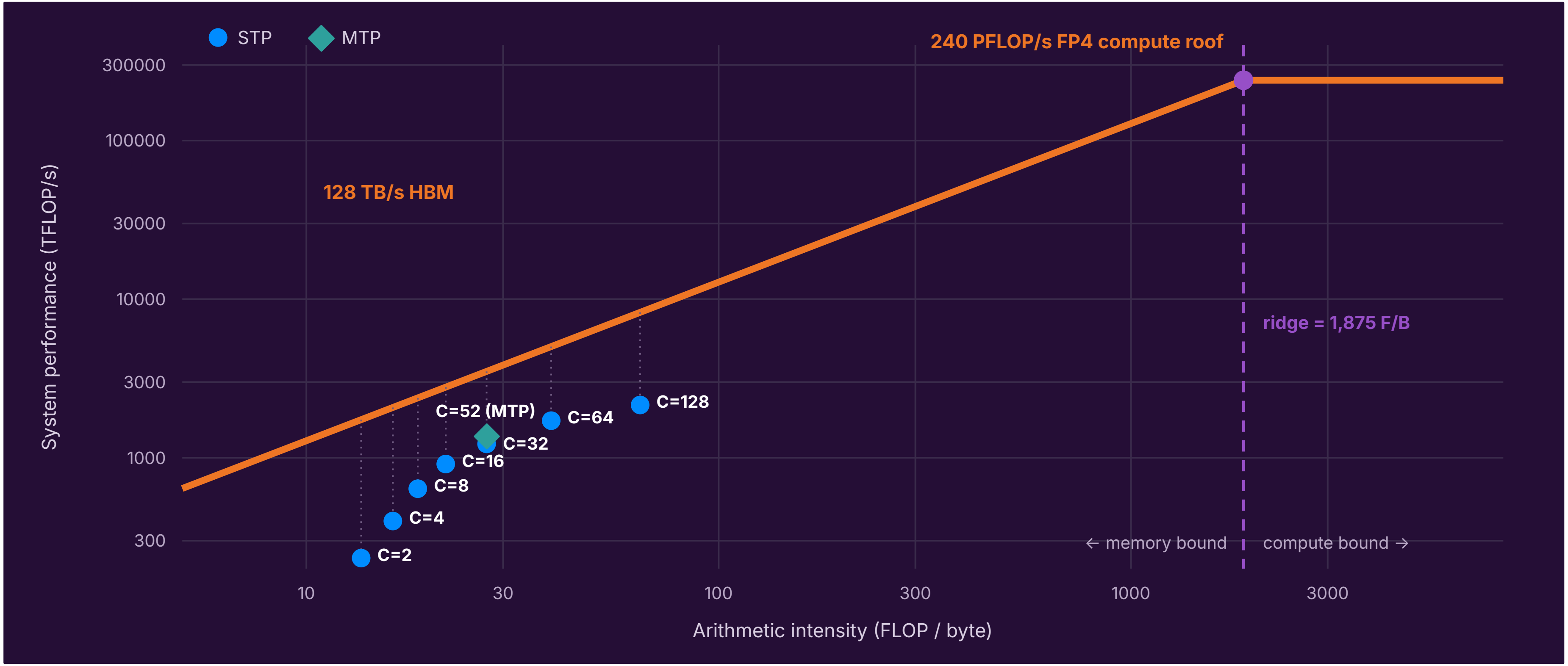
All AgentX frontier models; DeepSeek-V3 8K/1K reference included



AgentX, Cao4, et. al.; decode = (OSL-1) * TPOT. DeepSeek-V3 reference: 8K input / 1K output

Decode is Bandwidth-Bound

DeepSeek-R1 8K/1K • B300 FP4 • InferenceX measured data • STP + MTP3 • 16 decode GPUs



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MTP3: InferenceX batch8/width4; NVIDIA R1 mean acceptance length 2.82 is used only to infer target-forward rate.

HBM Bandwidth Utilization ≠ Model Bandwidth Utilization (MBU)

MBU = fraction of peak HBM bandwidth used for weights and KV caches

$$\textbf{tokens/s} = \frac{\text{HBM bandwidth/chip} \times \text{MBU} \times \text{number of chips}}{\text{parameter bytes} + \text{KV-cache bytes/token}}$$

Peak HBM bandwidth
What the hardware can move

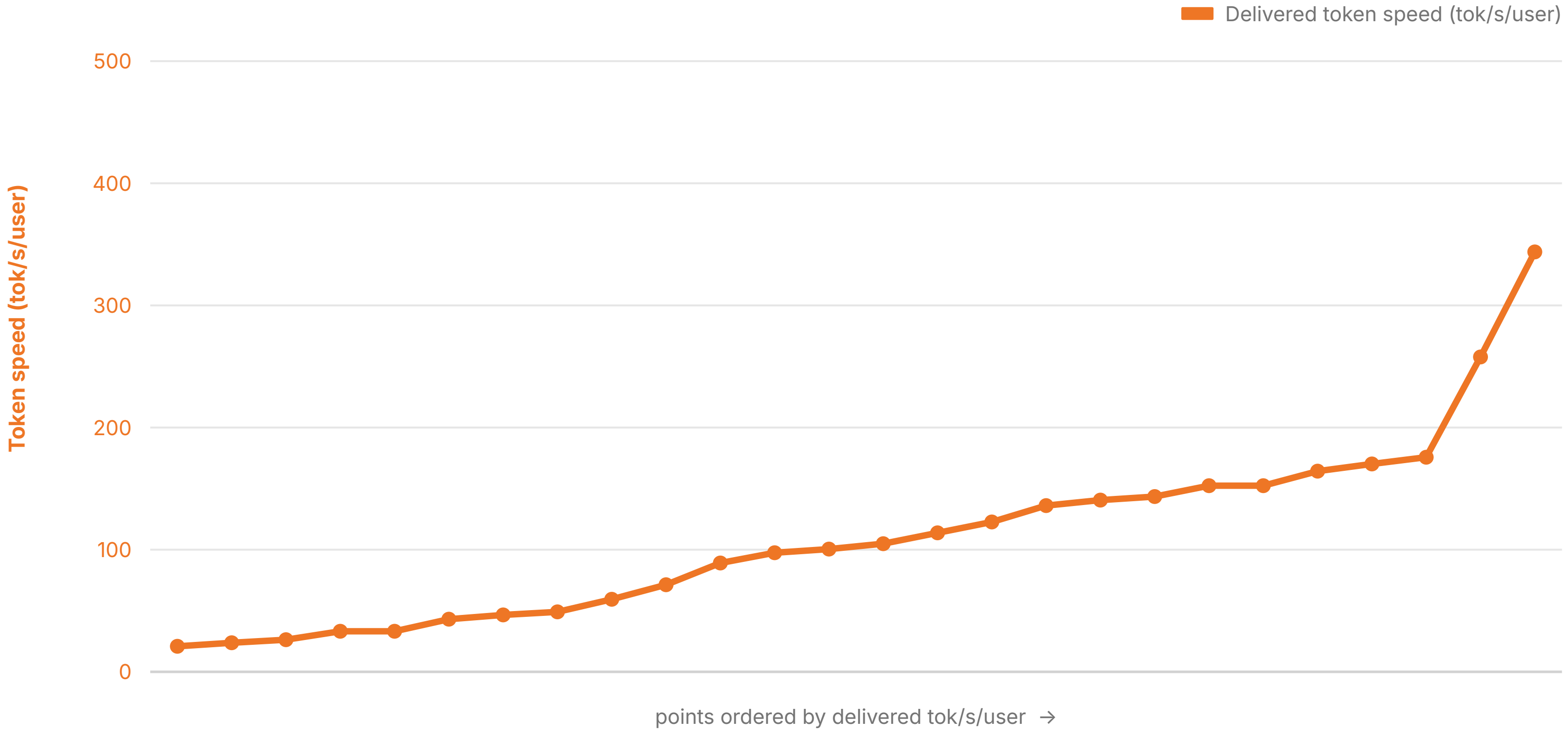
MBU
What the model actually uses of it

Token speed
Follows MBU, not just chip count

Reference: Databricks — LLM Inference Performance Engineering: Best Practices — databricks.com/blog/llm-inference-performance-engineering-best-practices

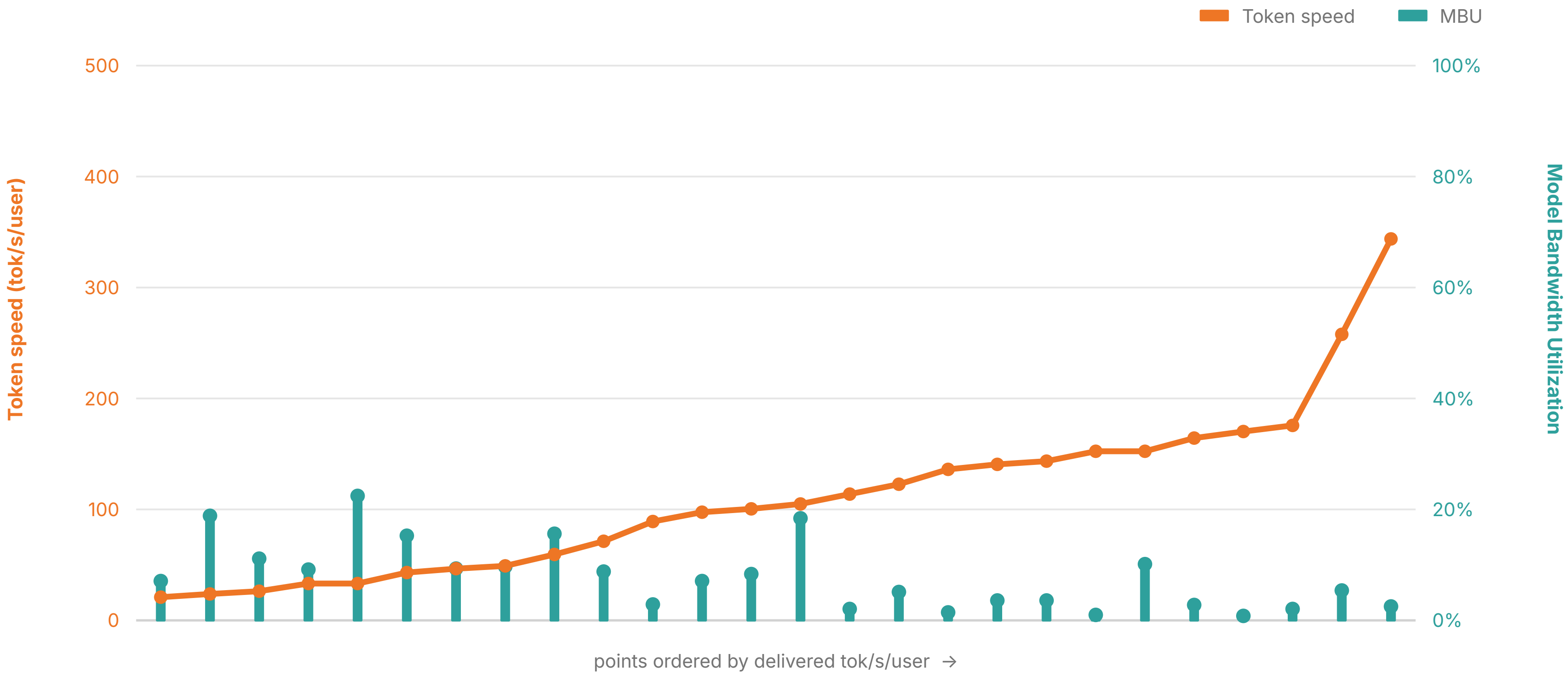
What the world sees: higher token speed

DeepSeek-R1 8K/1K | B300 FP4 | all 26 InferenceX points | sorted by delivered tok/s/user



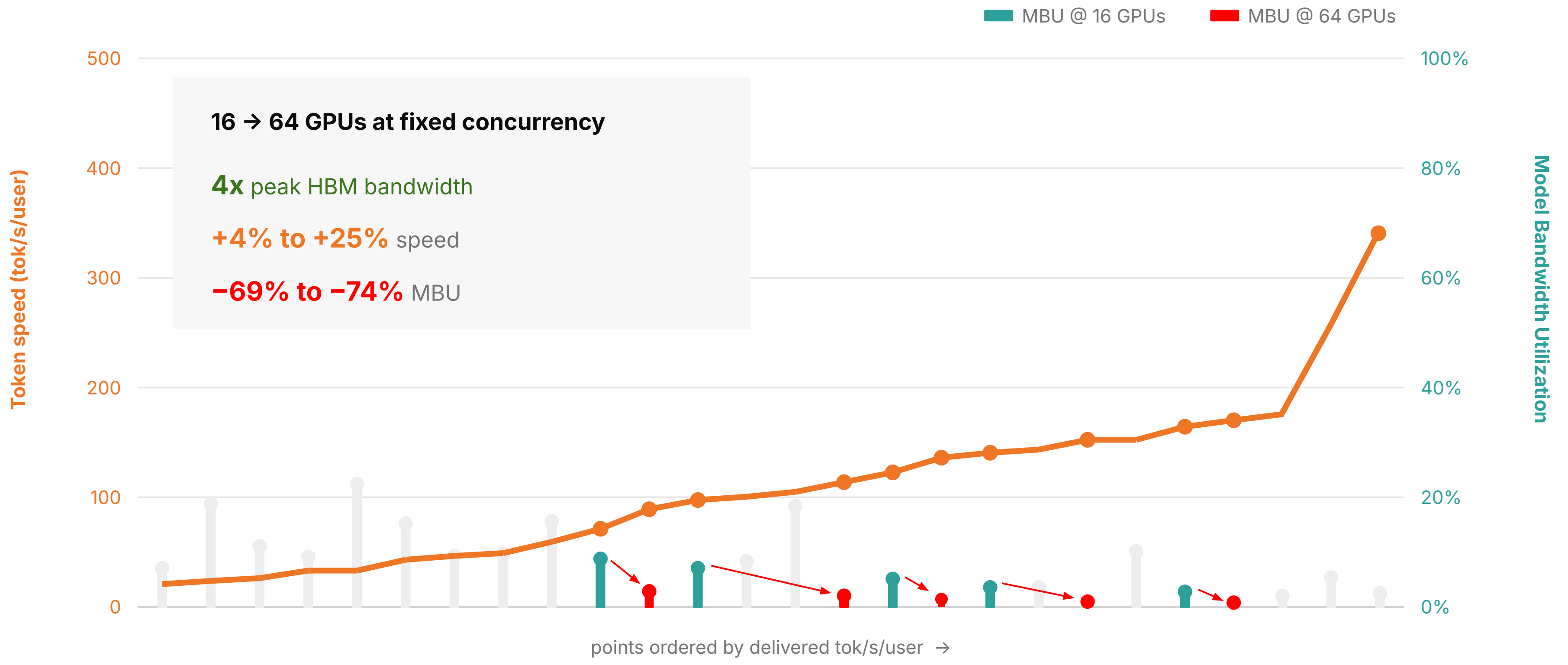
GPUs: Low Model Bandwidth Utilization

DeepSeek-R1 8K/1K | B300 FP4 | all 26 InferenceX points | sorted by delivered tok/s/user



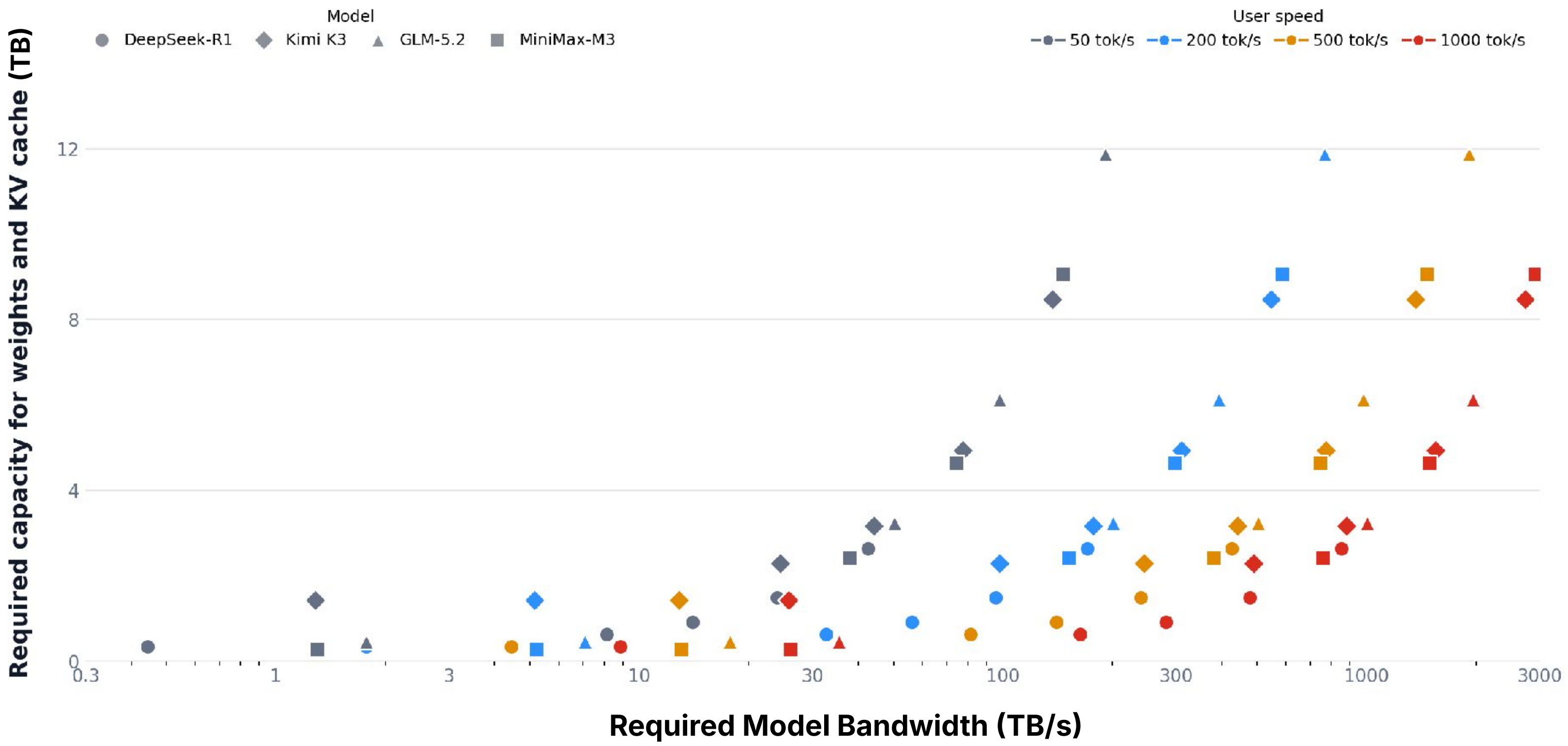
Increasing GPUs from 16 → 64: Higher Speed, Lower MBU

DeepSeek-R1 8K/1K | B300 FP4 | 26 InferenceX GPU points | sorted by delivered tok/s/user



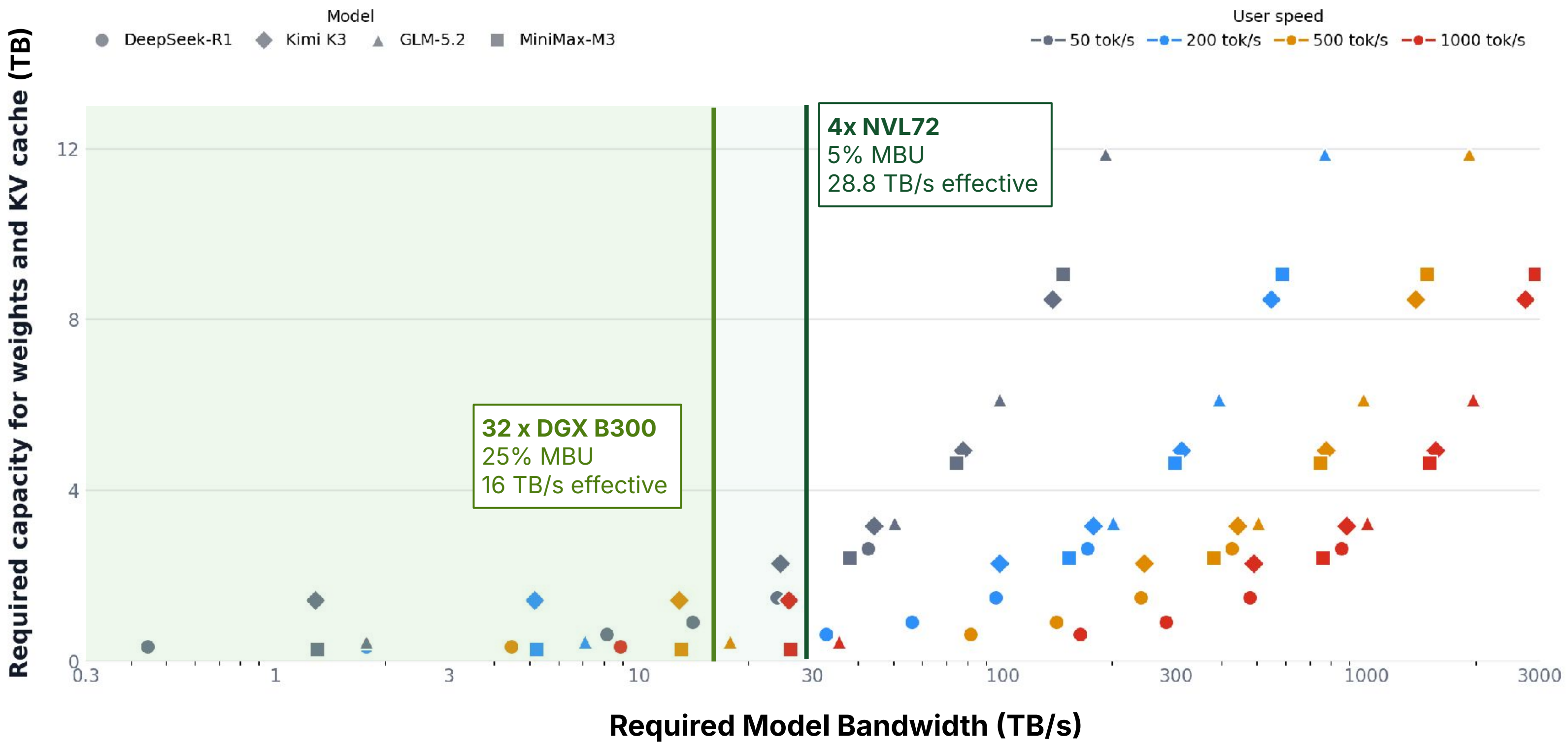
Capacity and MBU: Frontier Models Demand Scale

DeepSeek-R1, Kimi K3, GLM-5.2, MiniMax-M3 | points only | MTP3 accepted length = 3.1



GPUs @500 kW: Power Buys Capacity, not Bandwidth

All frontier-model points | MTP3 accepted length = 3.1



Agentic Inference needs **High Bandwidth** at **Scale**

SN50: DataFlow Built For Scale

3200 TFLOPS (FP8)
5X SN40

2 TB/s Ethernet
Scale-up Network

256+ SN50
Scale-up Domain

400G
Scale-out Network

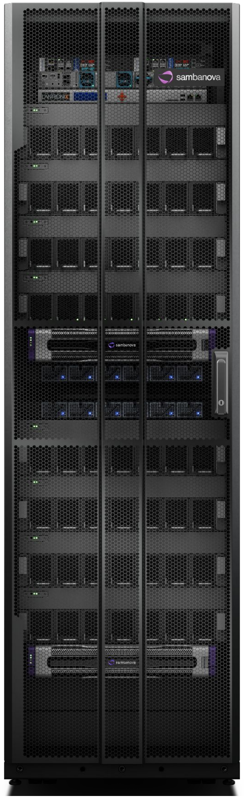
5nm CoWoS-L

432 MB On-Chip SRAM

64 GB, 1.84 TB/s High
Bandwidth Memory

CXL-based DDR for
Prompt Caching, 100 GB/s

SN50 Air-Cooled Rack Architecture



1G Switch Console

AC PDUs

node1 XRDU
<0,1,2,3>

Host1

Power Shelves

node0 XRDU
<0,1,2,3>

Host0

Total per Rack: 16 RDUs, 2 Nodes

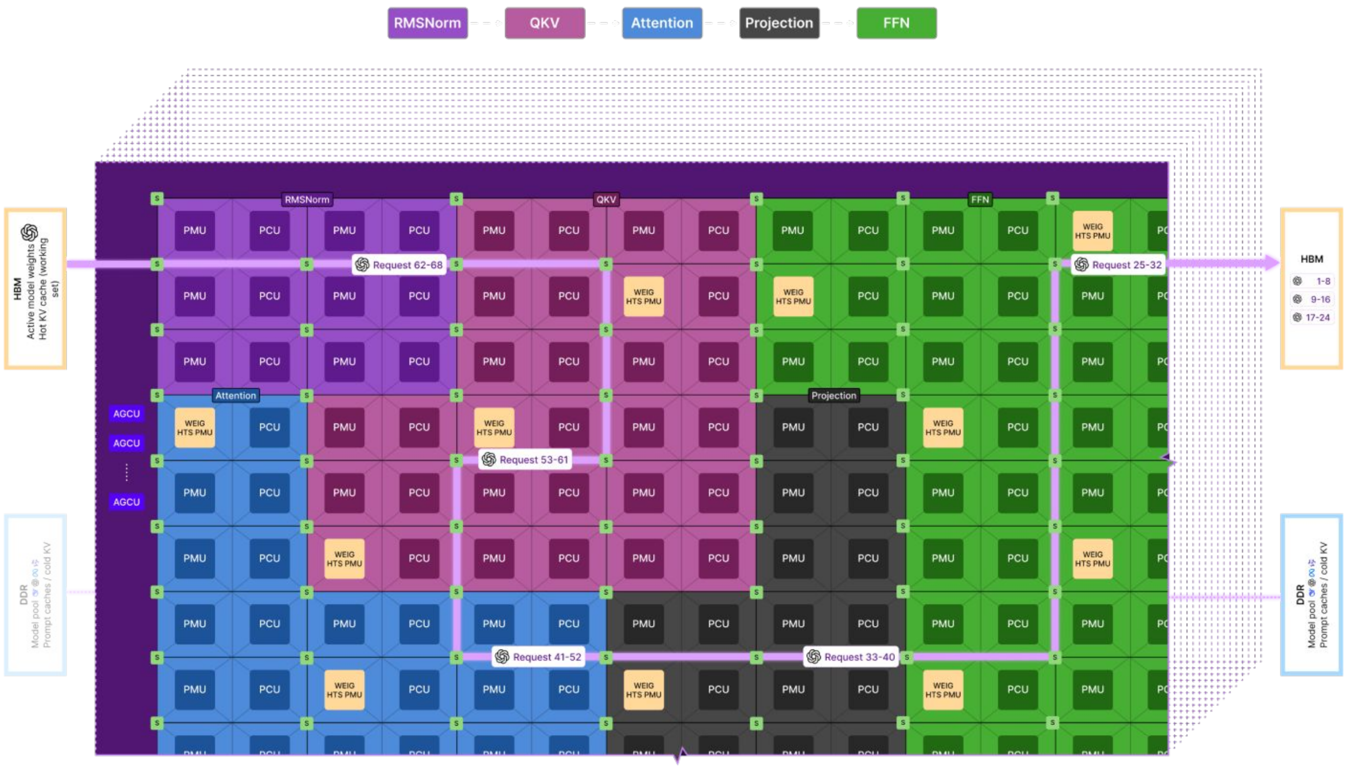
- Compute: 25.6 PFLOPs @BF16, 51.2 PFLOPS @FP8
- SRAM: 6.9 GB
- HBM: 1 TB
- RDU DDR: 256 GB to 2 TB
- Host: 2× 64-core CPUs, 1 TB DDR5
- Host boot and storage: 4× 960 GB NVMe SSD
- Host storage: 6× 7.6 TB NVMe SSD
- Networking
  - Scale-up: 6.4 TB/s
  - Scale-out: 800 GB/s
  - Host: 4×100G
- Management: 1 GbE switch / serial console server
- Power Consumption:
  - Typical: 15 - 20 kW
  - Maximum: Up to 34 kW
- Cooling: Air-cooled
- Rack Dimensions:
  - Height 2026 mm, Width 600mm, Depth 1272mm (including doors)

SN50: **Dataflow** Built For Scale

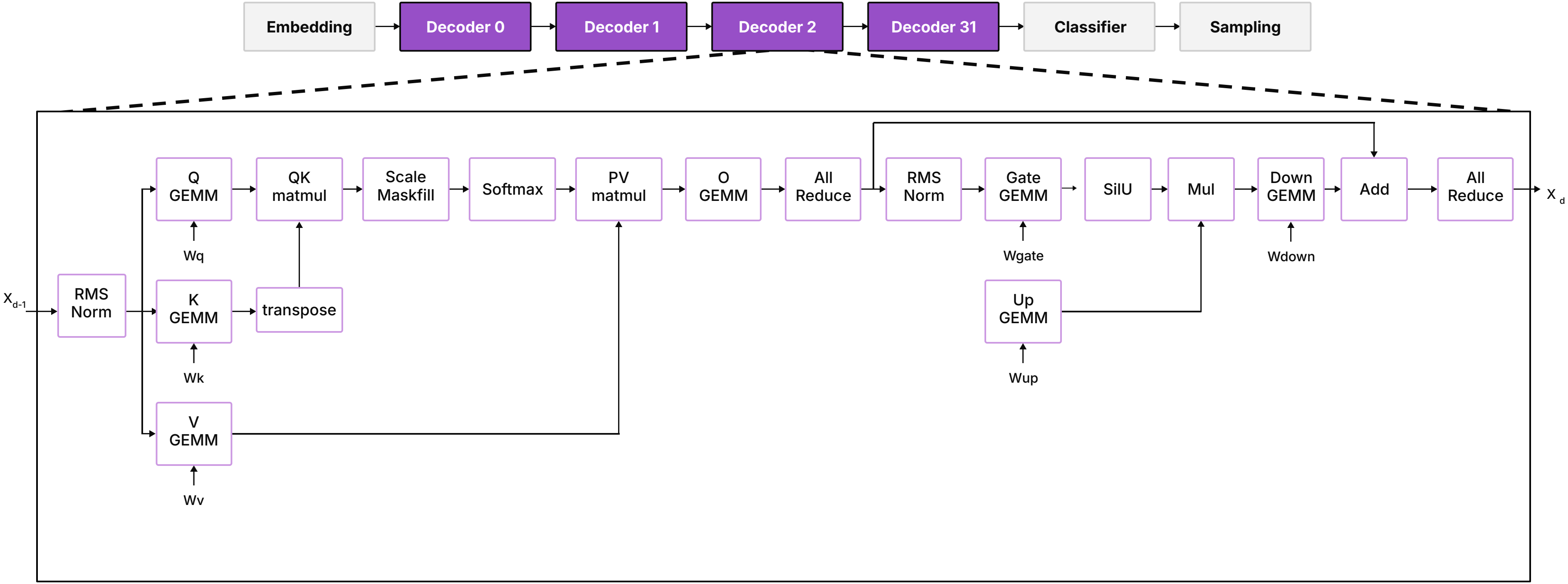
SN50: Chip Architecture

864 PCUs and PMUs

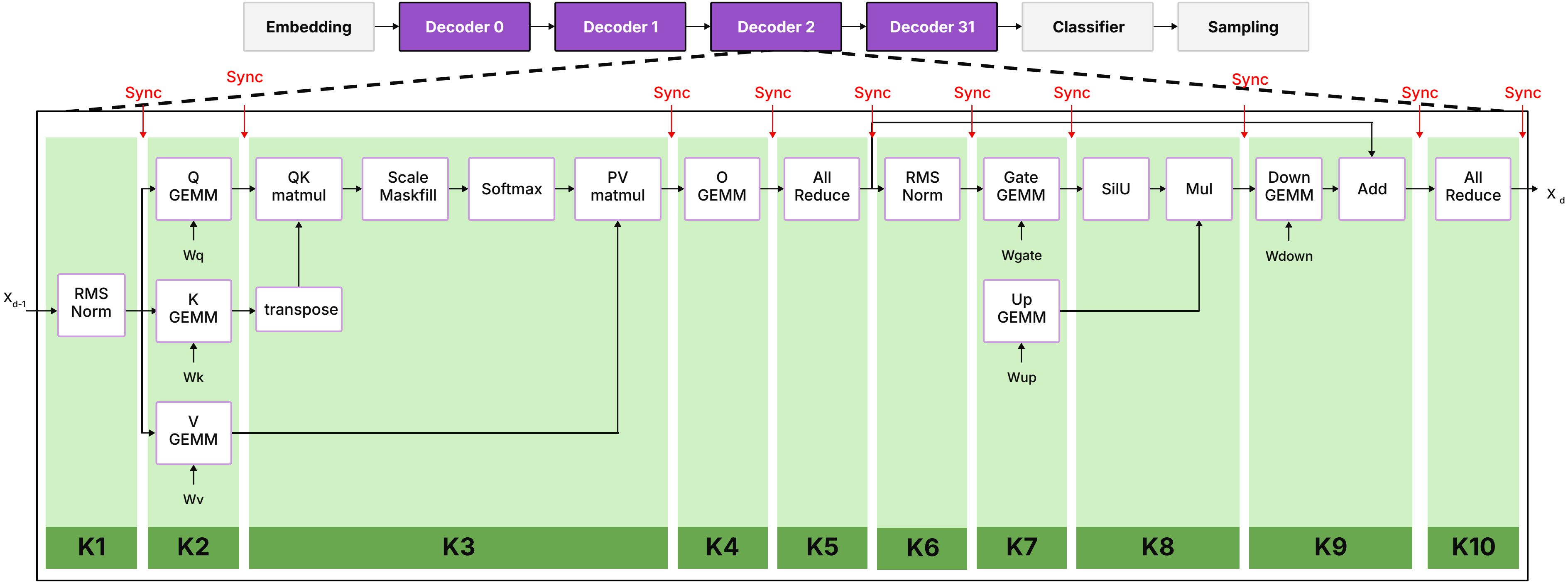
PCU: Compute unit
PMU: Memory unit
S: Mesh switches
AGCU: Portal to off-chip memory and I/O



Structure of a Transformer

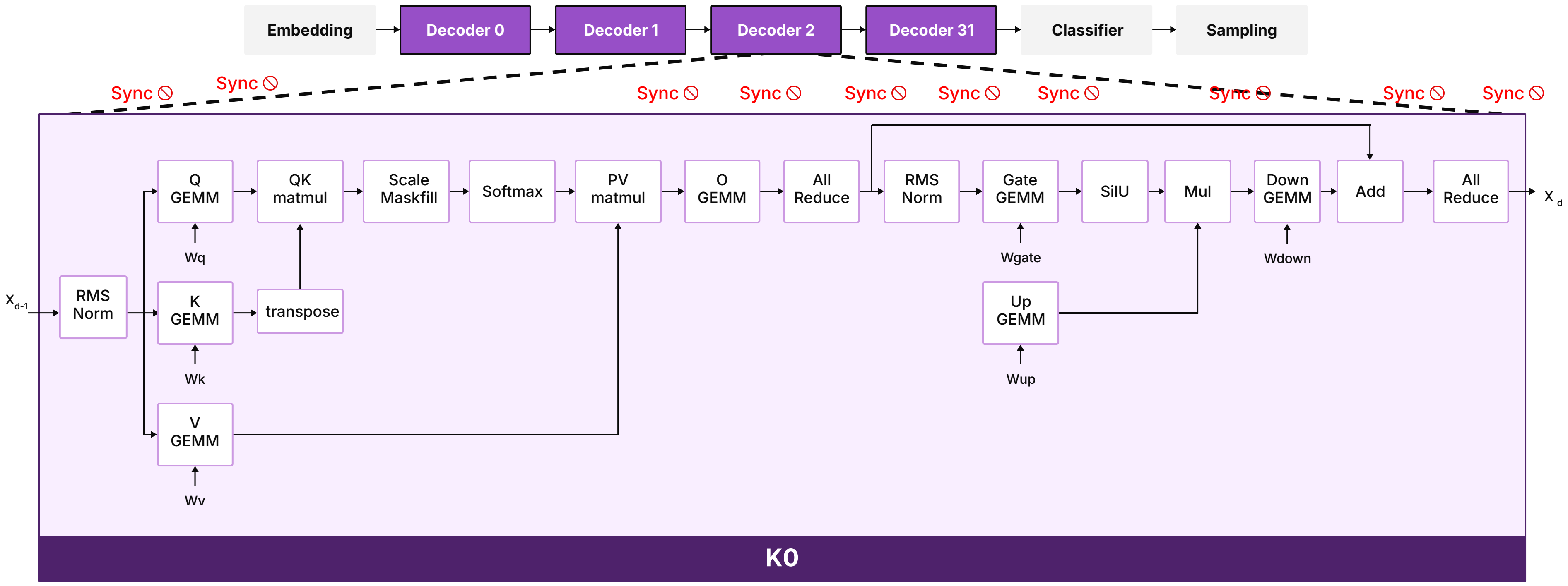


GPUs Incur Overheads with Lower Operator Fusion



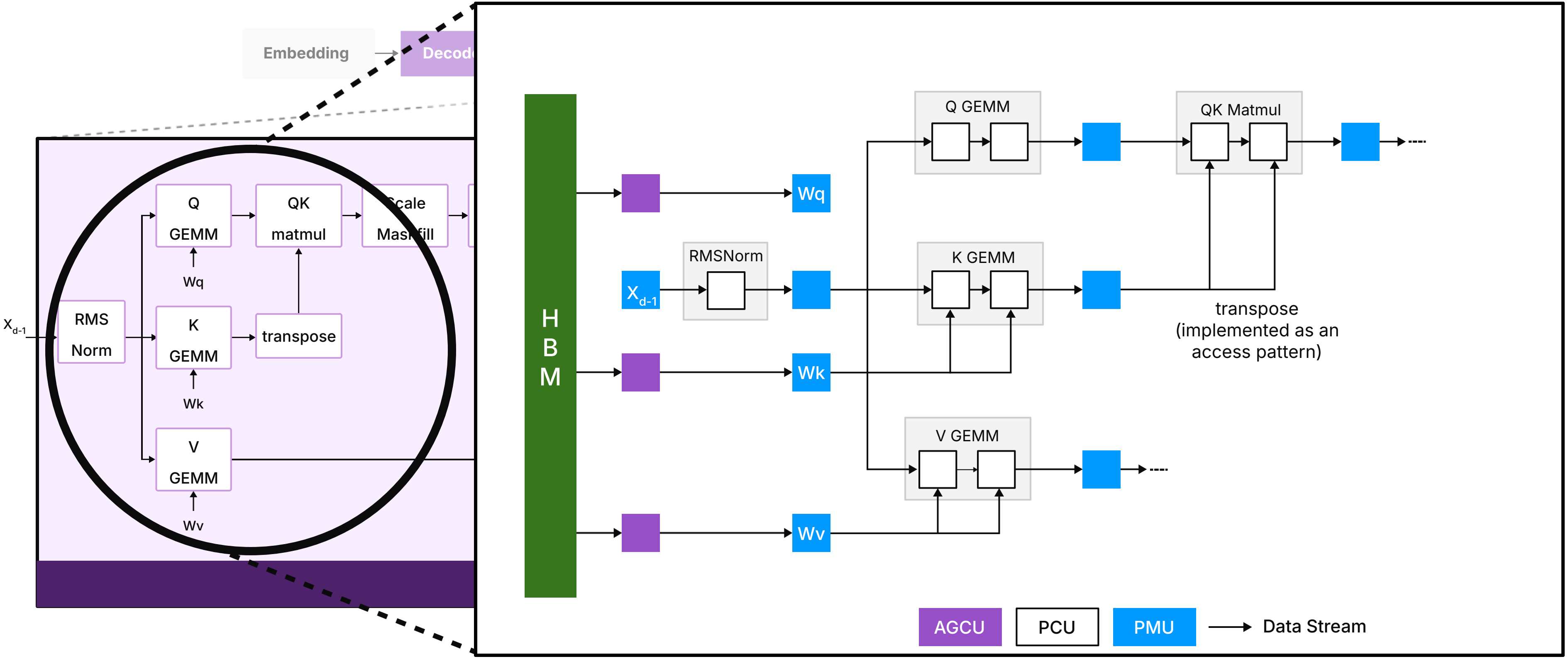
Low Operator Fusion **Low** Data Locality **High** Launch and Synchronization Overheads

SN50: Persistent Decoder Kernel, No Overheads!

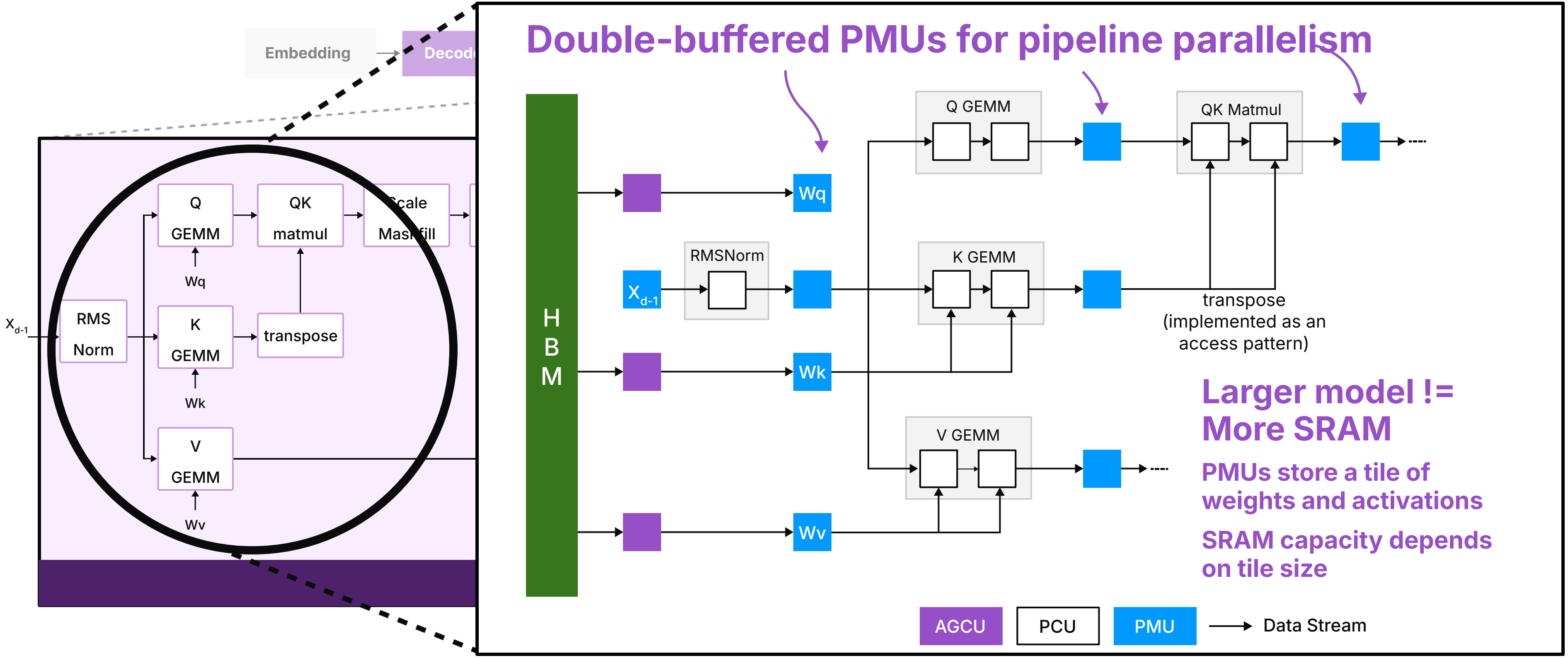


High Operator Fusion: One Kernel Call for All Decoders! **Zero Kernel Launch Overheads** High Data Locality

Compute - Communication Overlap: Within RDU



Compute - Communication Overlap: Within RDU

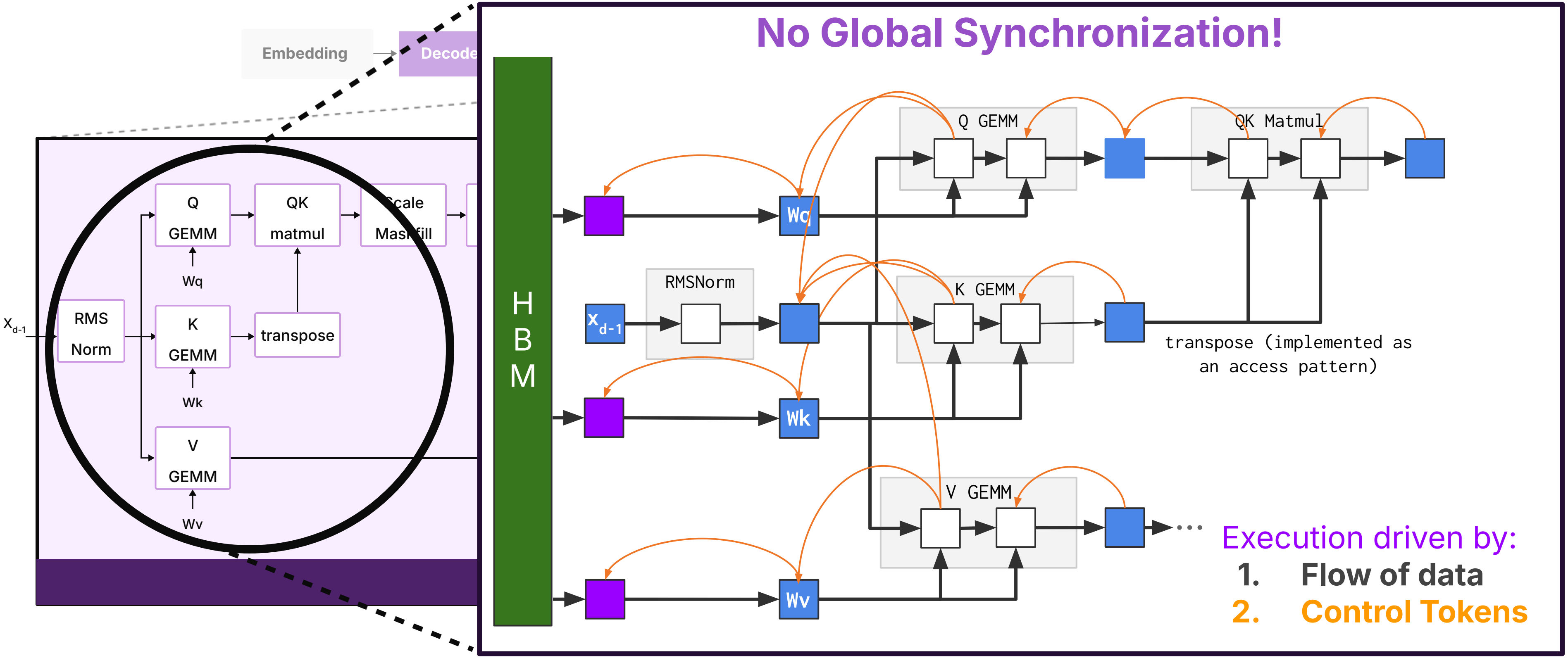


Larger model != More SRAM

PMUs store a tile of weights and activations

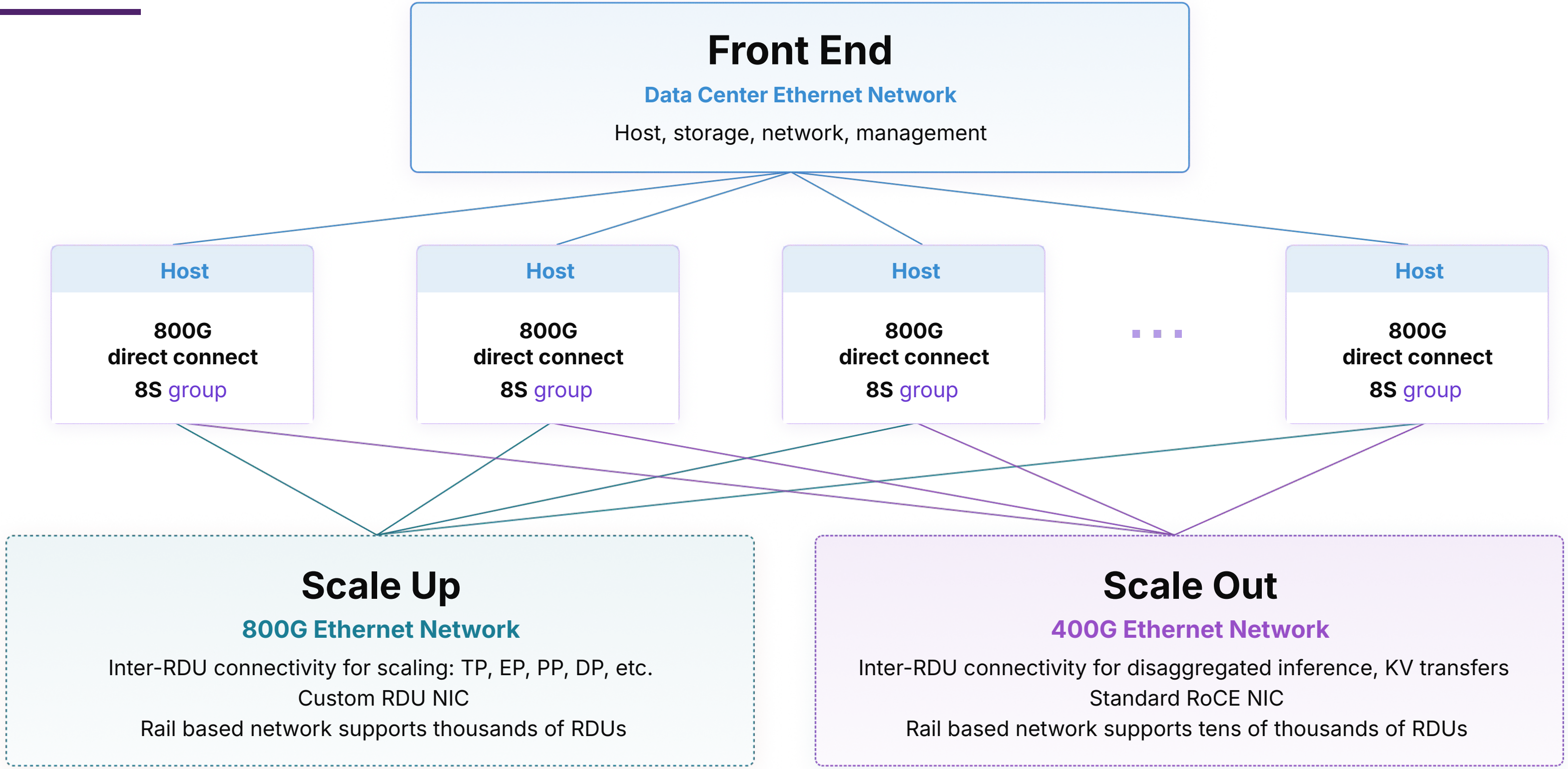
SRAM capacity depends on tile size

Compute - Communication Overlap: Within RDU



SN50: Dataflow **Built For Scale**

SN50 Scaling



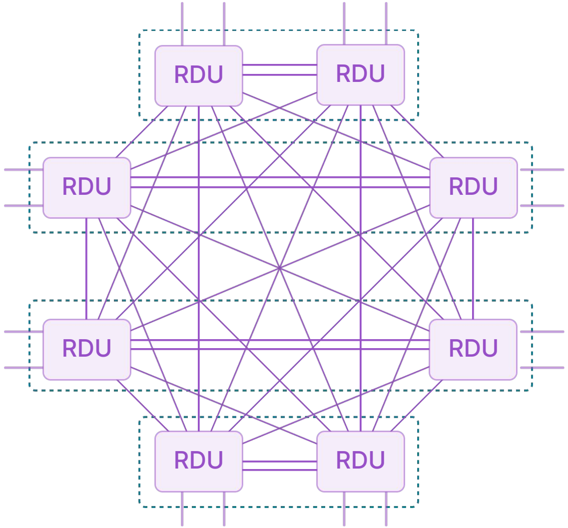
Performant, cost effective scaling: fully connected intra-node, scale up network and scale out network enable solutions for a broad range of system applications

SN50 Node: Scale-Up and Scale-Out

- RDUs have 10x integrated 800G Ethernet ports for scaling; 2 TB/s per RDU
  - 7 ports for fully connected 8S node scale up; 2 ports for scale up network
  - 1x RoCEv2 NIC per SN50 RDU used for scale out between scale up domains

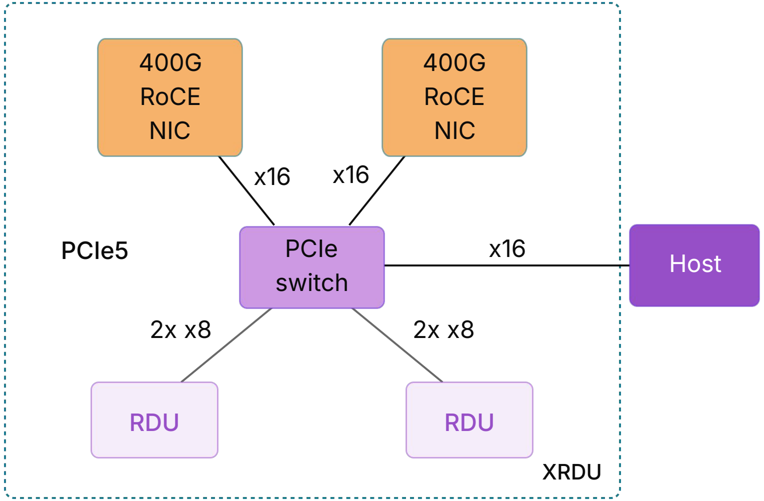
8-socket Node: Scale-Up Topology

400 GB/s per RDU for inter-node scale up



200 GBps, 800G Enet links

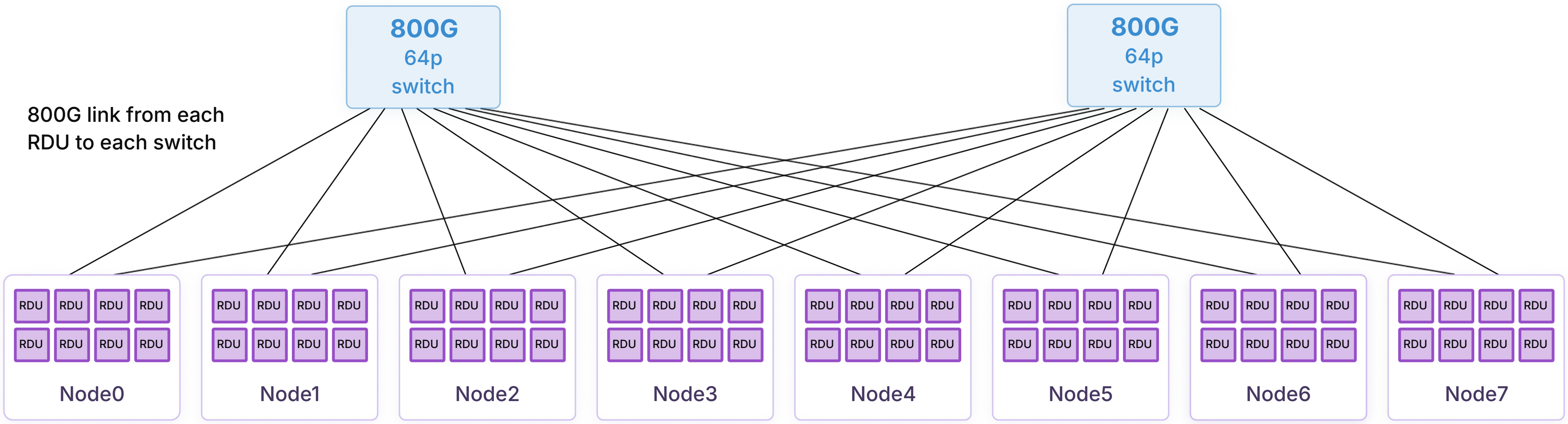
8-socket Node: 1 Scale-Out NIC per RDU



50 GBps Enet per RDU with one 400G NIC per RDU

Scaling Up Beyond One Node: 64-socket Example

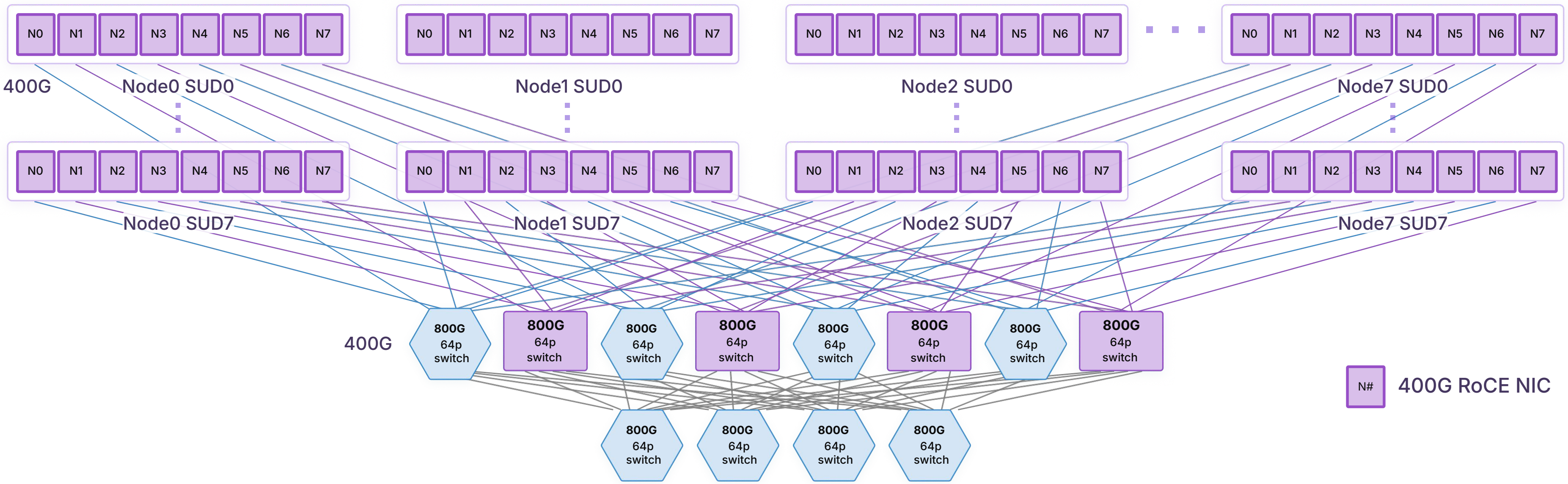
- 64S SN50 scale up can be achieved with 2× 64p 800G switches
  - Each of the 64 RDUs connects one 800G link to each switch



High bandwidth, single tier switching architecture for 64 sockets

Scaling Out across Scale-Up Domains: 512 Socket Example

- Rail optimized 2-tier network
  - 1× 400G RoCE NIC per SN50 RDU
- 512S scale out domain with 12× 800G 64p switches



Simple system scale out network

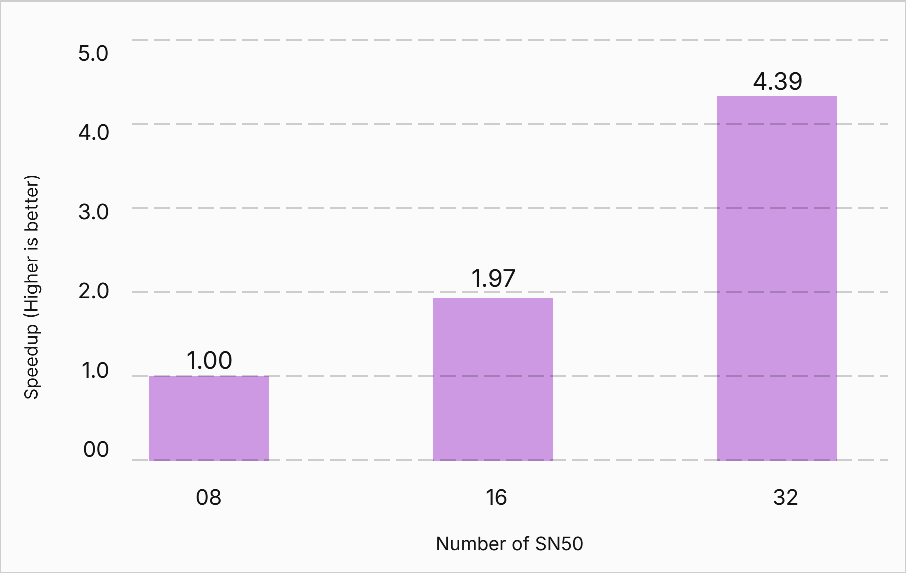
SN50: **Dataflow** meets **Model Parallelism**

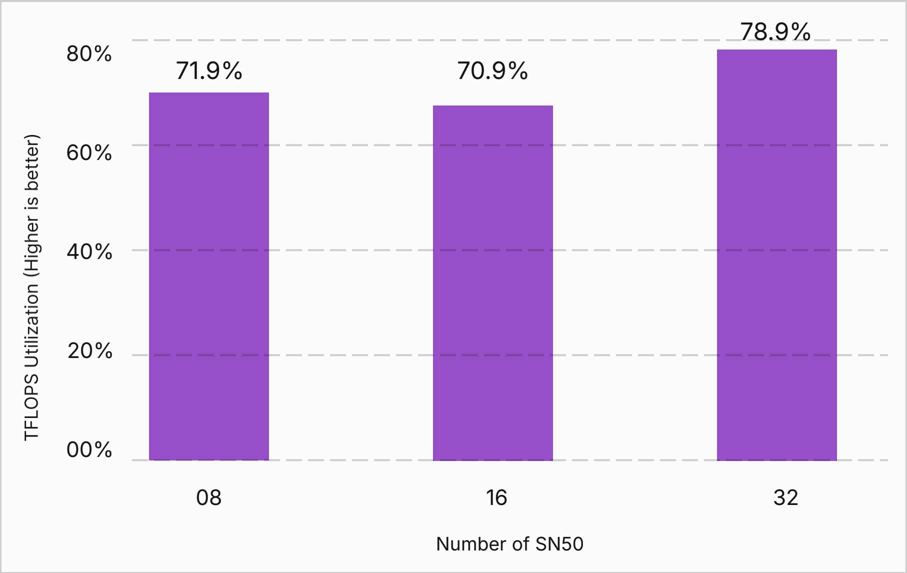
SN50 Supports All Forms Of Model Parallelism

Model Parallelism		Communication Primitives				
Tensor Parallel (TP)	→	Reduce-Scatter (RS)	+	All-Gather(AG)	OR	All-Reduce(AR)
Pipeline Parallel (PP)	→	Send-Receive				
Expert Parallel (EP)	→	All-to-All				
Data Parallel (DP)	→	Reduce-Scatter (RS)	+	All-Gather(AG)	OR	All-Reduce(AR)

Tensor Parallel and Data Parallel resolve to the same collective pair
the same fabric path serves both, so scaling one does not require a second interconnect strategy.

TP: SN50 On-Silicon Tensor Parallel GEMM Benchmarking





Key takeaways:

- 70% or higher TFLOPs utilization consistently measured across 8, 16, and 32 sockets
- 32-SN50 achieves full overlap of compute and communication

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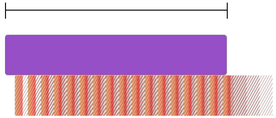
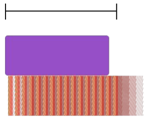
Importance of Overlap - First Principles

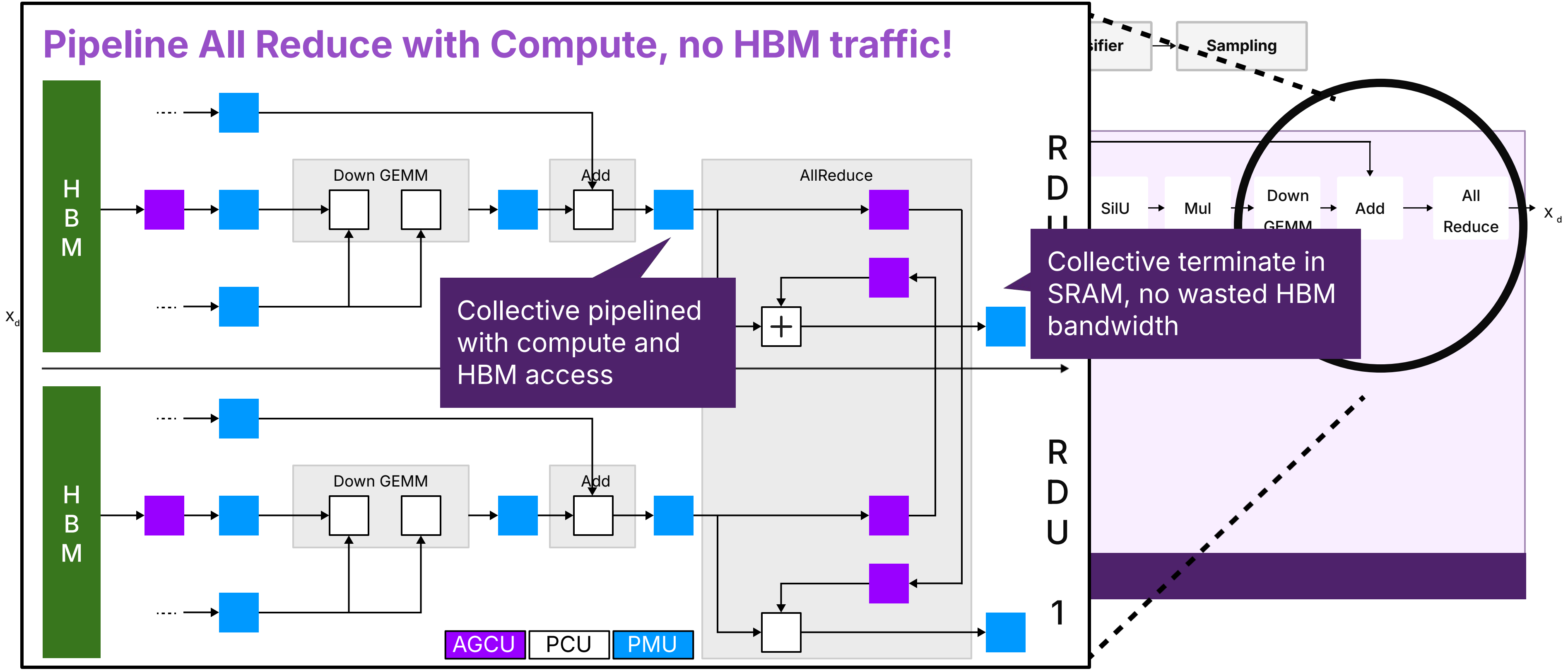
Compute Communication

Scale-Up Domain Size	Without Overlap (GPU)	With Overlap (RDU)
8 sockets		
16 sockets		
32 sockets		

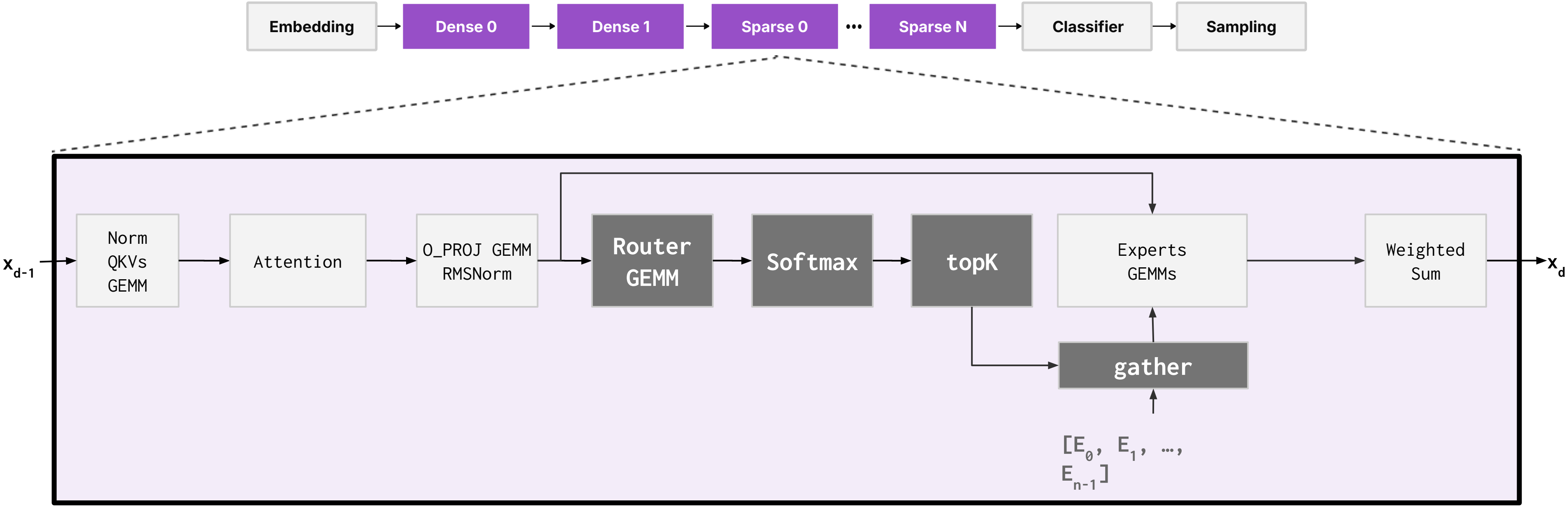
Separate communication phases create bottlenecks and extend total time.

Overlapped communication keeps work flowing and reduces total time.

TP: HBM and Compute Overlapped with Communication

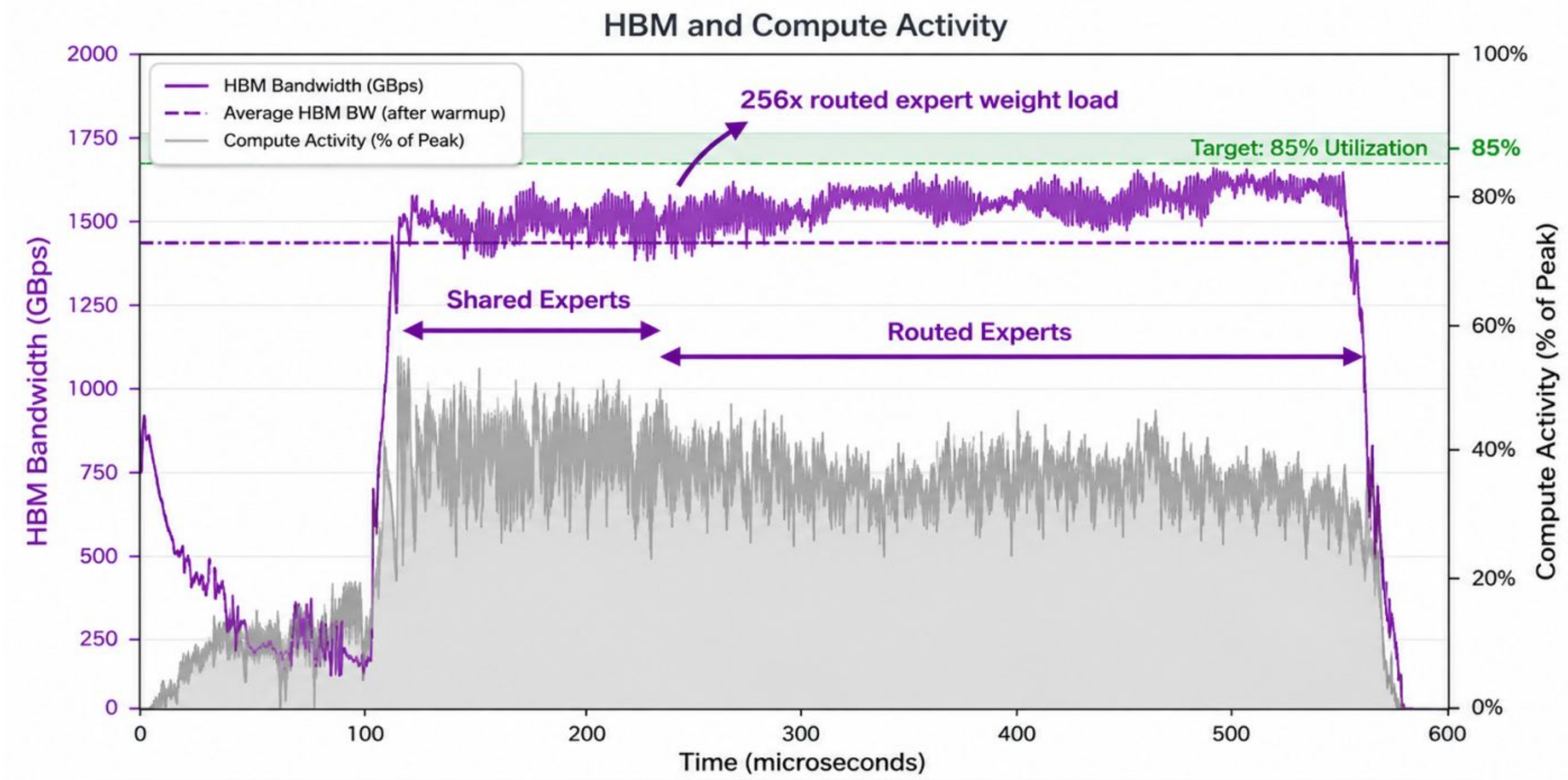


MoEs With TP

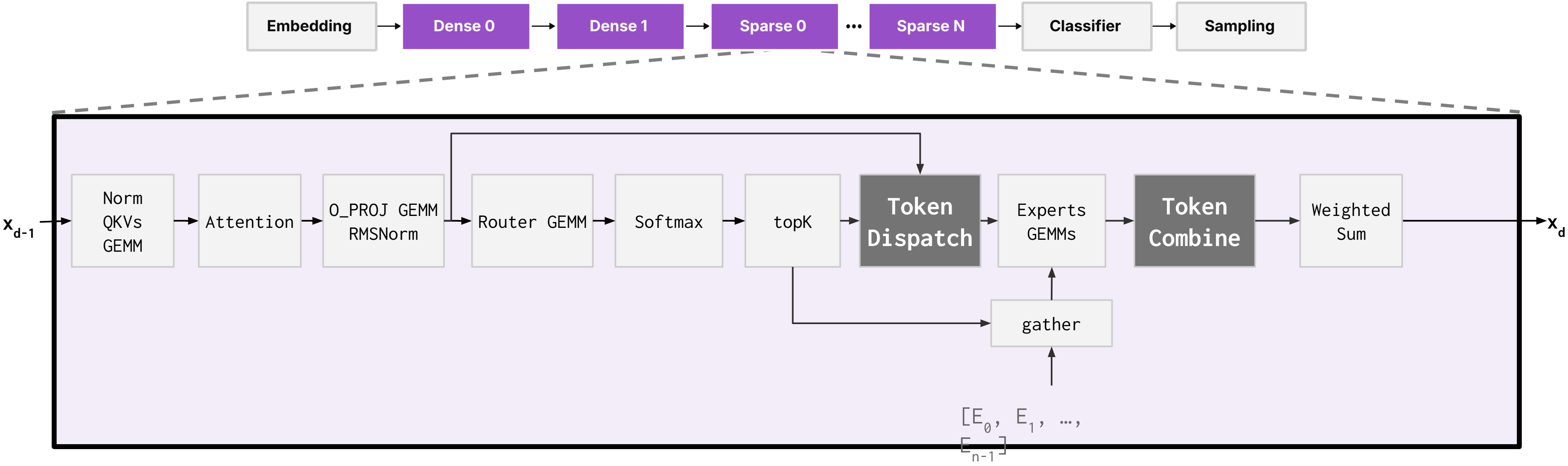


- MoEs are frequently mapped with TP-8, TP-16, and TP-32 on RDUs
- Kernel Looping extends to MoEs naturally with indexed weight loads
- Indexed loads work with both SRAM and off-chip memories (HBM, DRAM)
- To capture locality, experts of one layer are frequently prefetched into SRAM and indexed from there

DeepSeek TP-32 MoE Layers Achieve >70% MBU

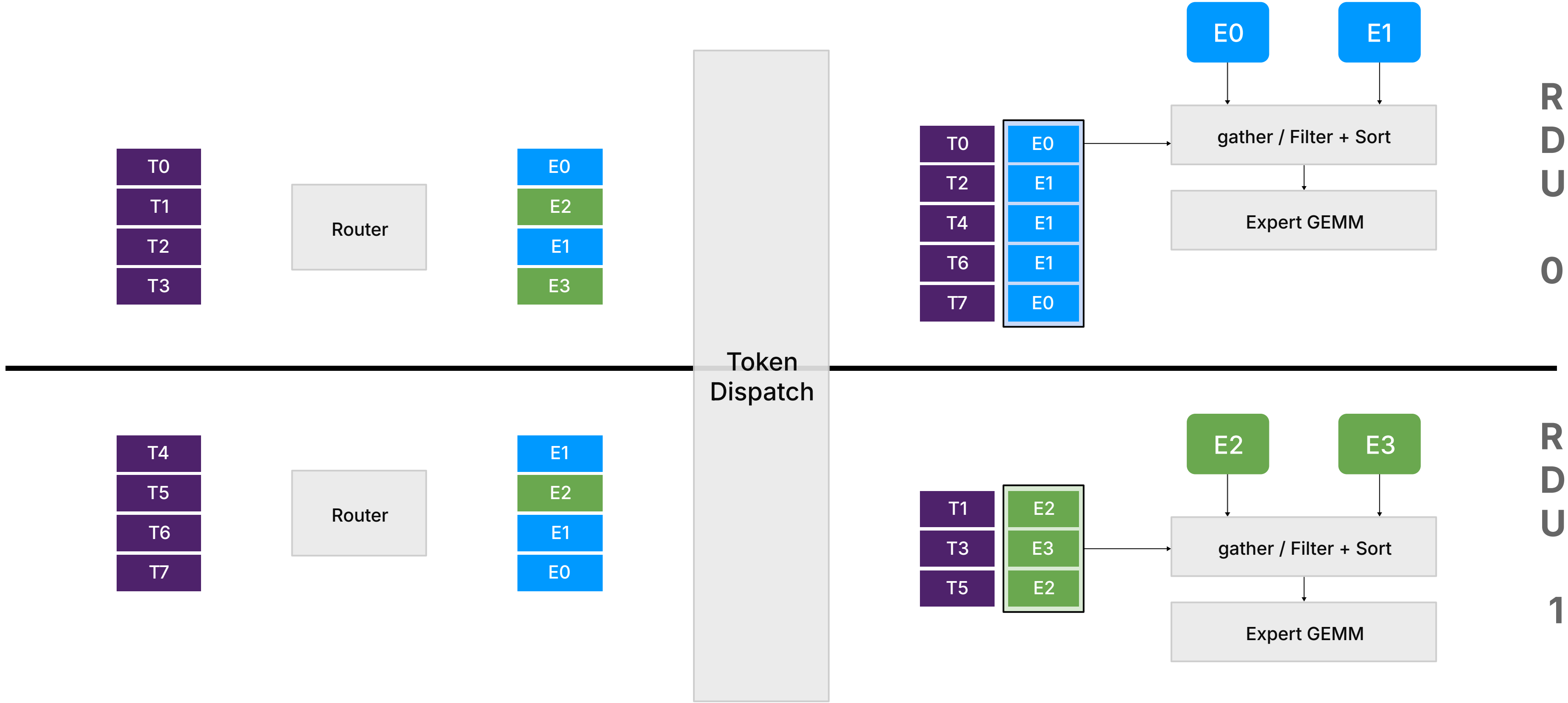


MoEs With TP **+ EP**



- EP introduces dynamic all-to-all communication across RDUs in **token dispatch** and **token combine**
- RDUs support two forms of token dispatch & combine: **broadcast, and all-to-all**
- Broadcast dispatch-combine: Uniform network traffic, extra network traffic, filtering at destination
- All-to-all dispatch-combine: Non-uniform network traffic, no extra network traffic, source filtering

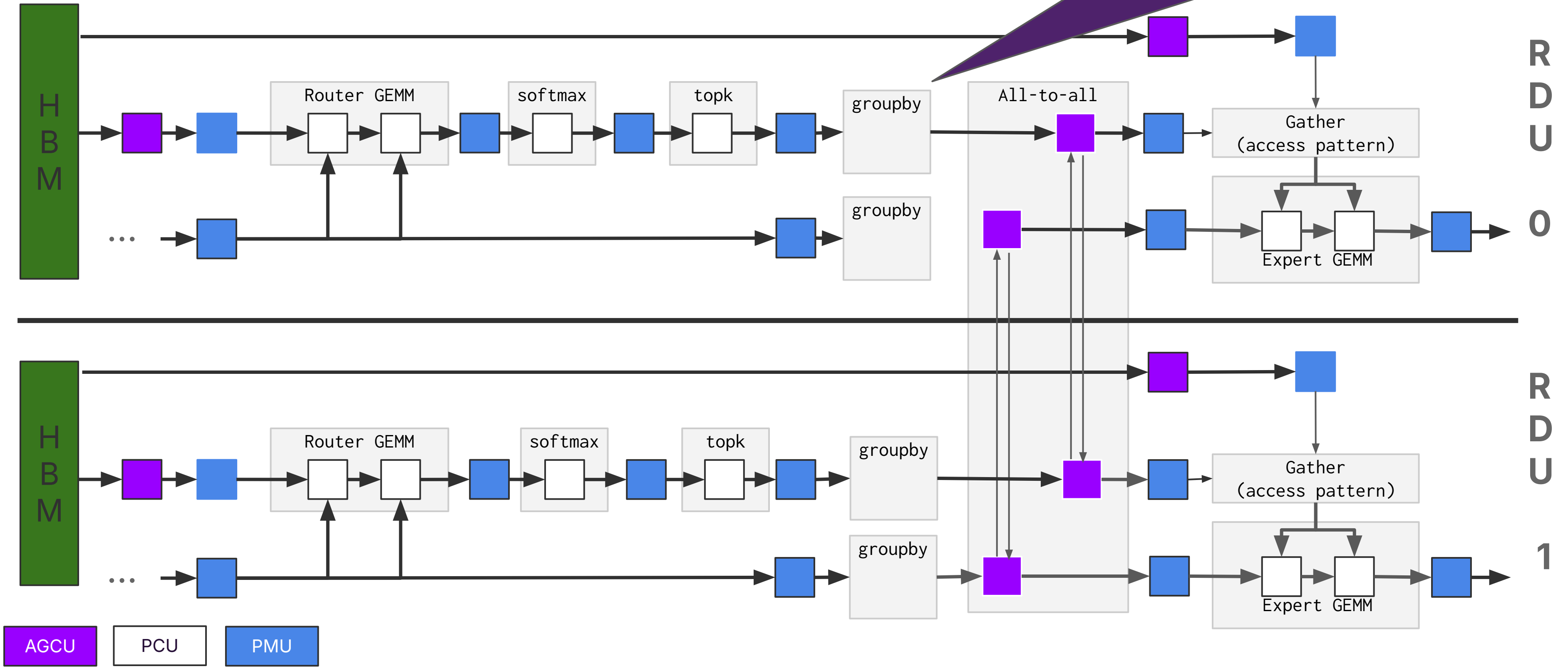
Token Dispatch: All-to-All



Token Dispatch: Broadcast + Filter

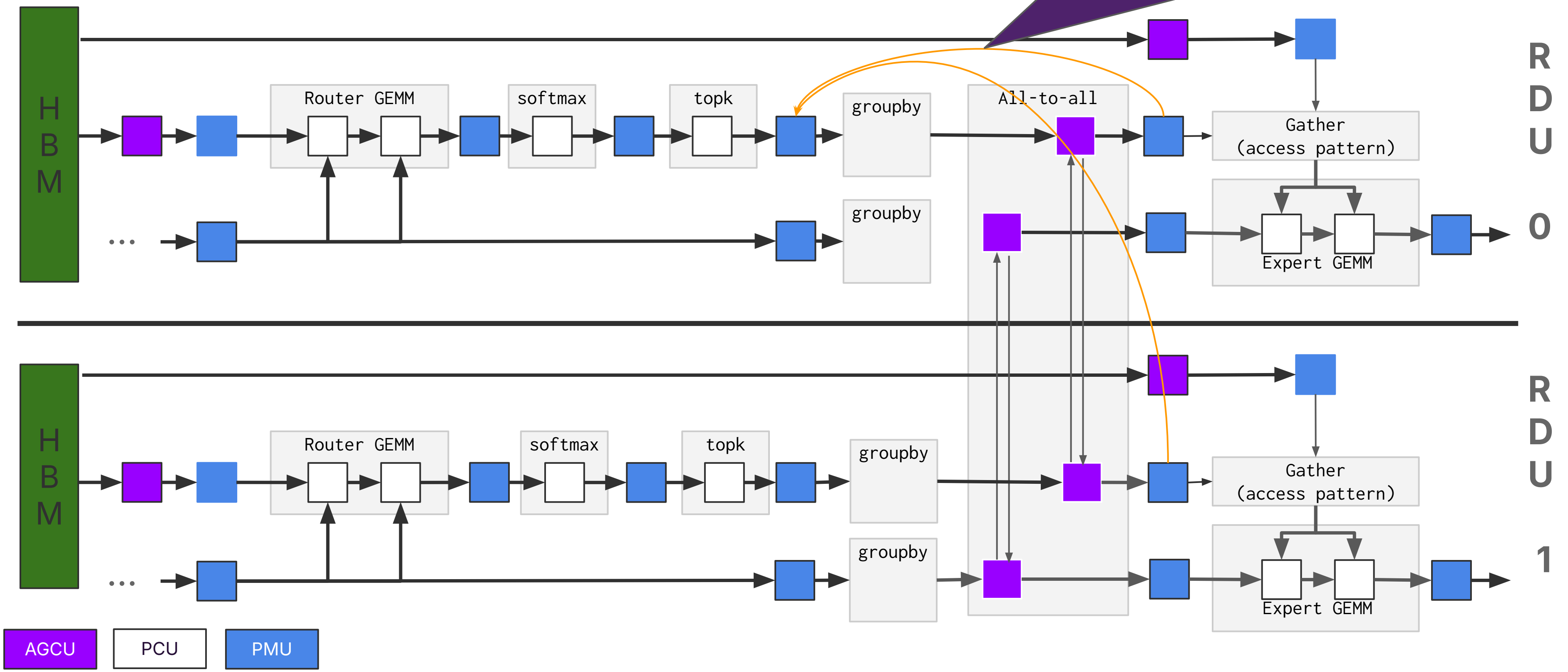
Token Dispatch: All-to-All

Group tokens and experts by bins, route to different RDU destinations

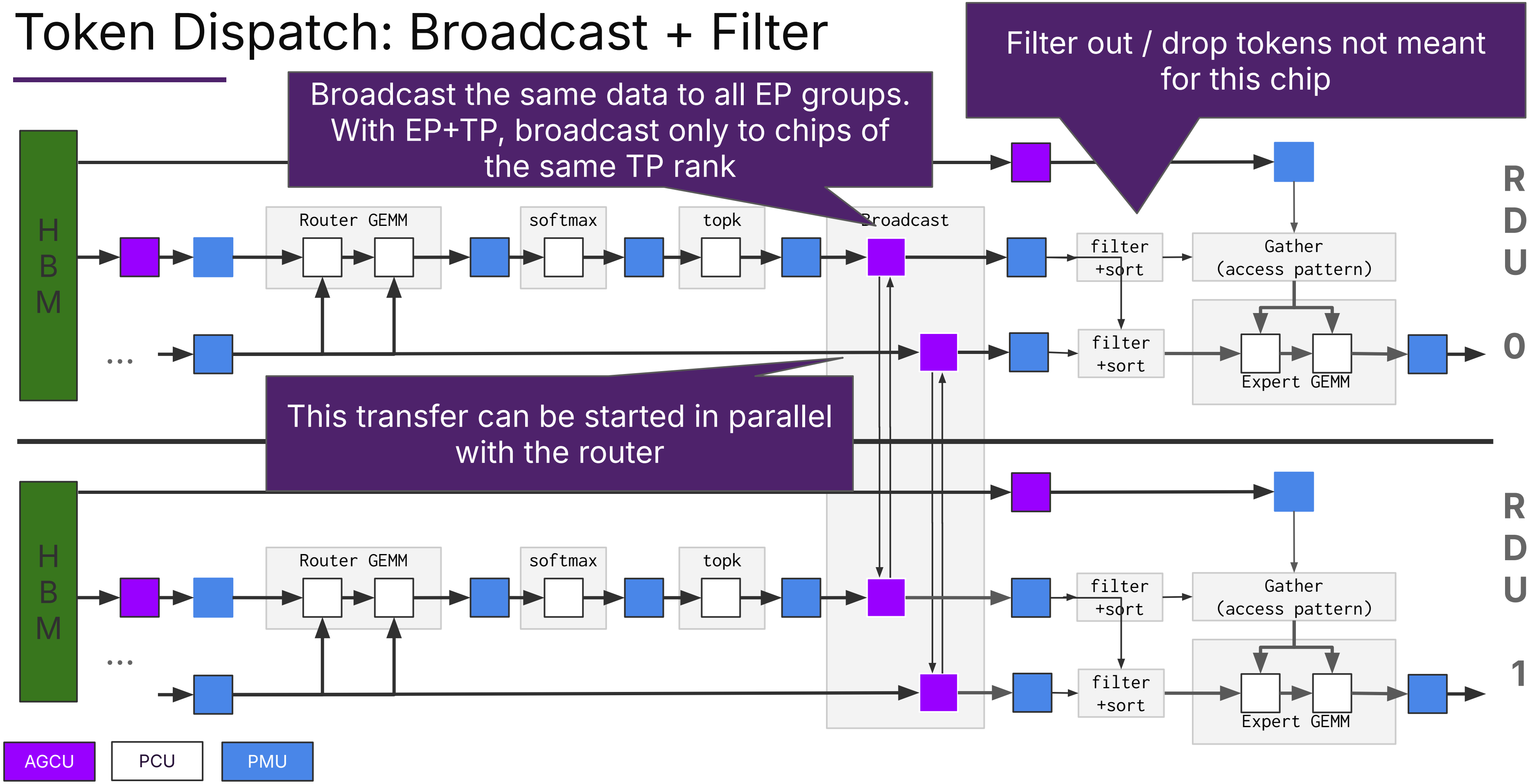


Token Dispatch: All-to-All

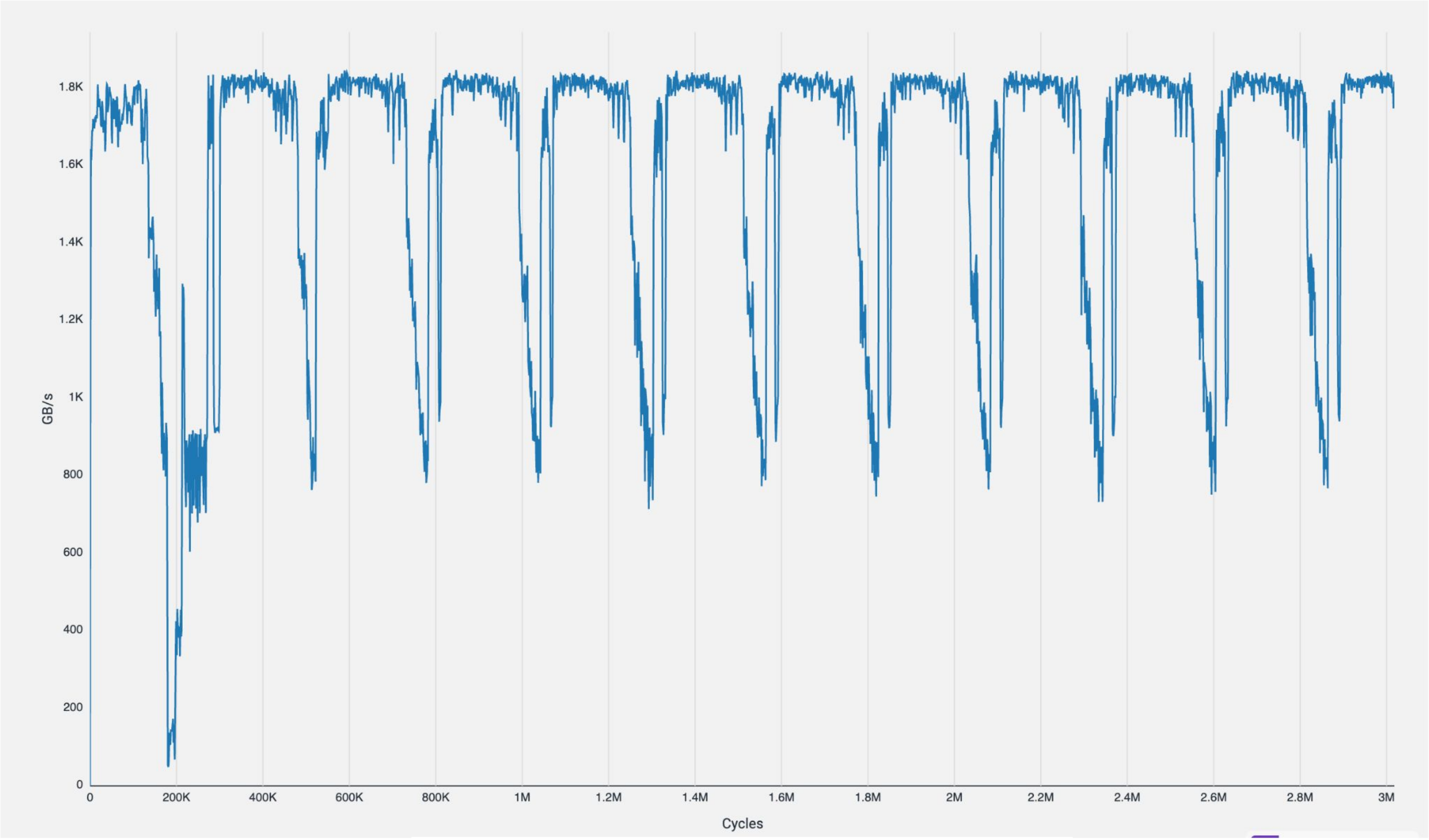
Independently flow-controlled streams:
Handles dynamism,
No buffer overflows



Token Dispatch: Broadcast + Filter



EP on 64-socket SN50: >70% Bandwidth To Load Experts



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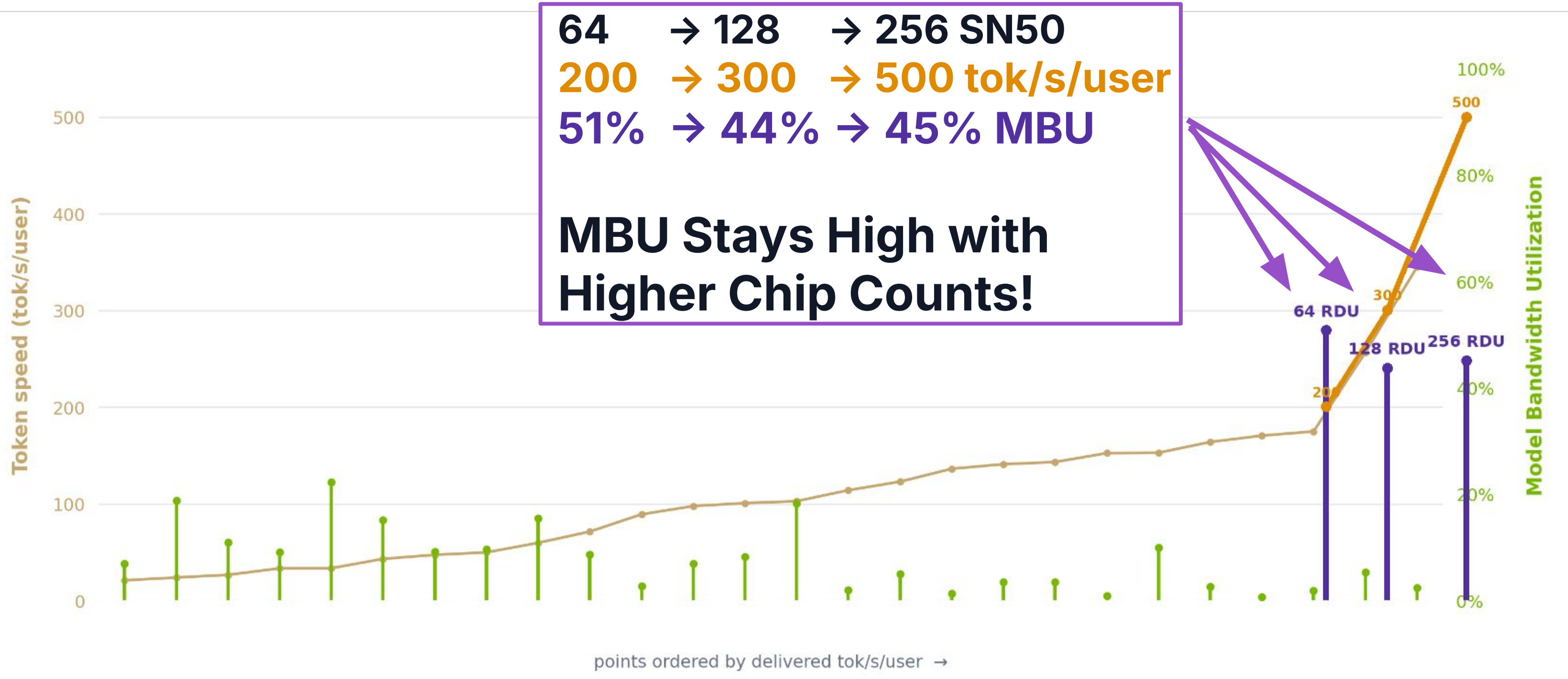
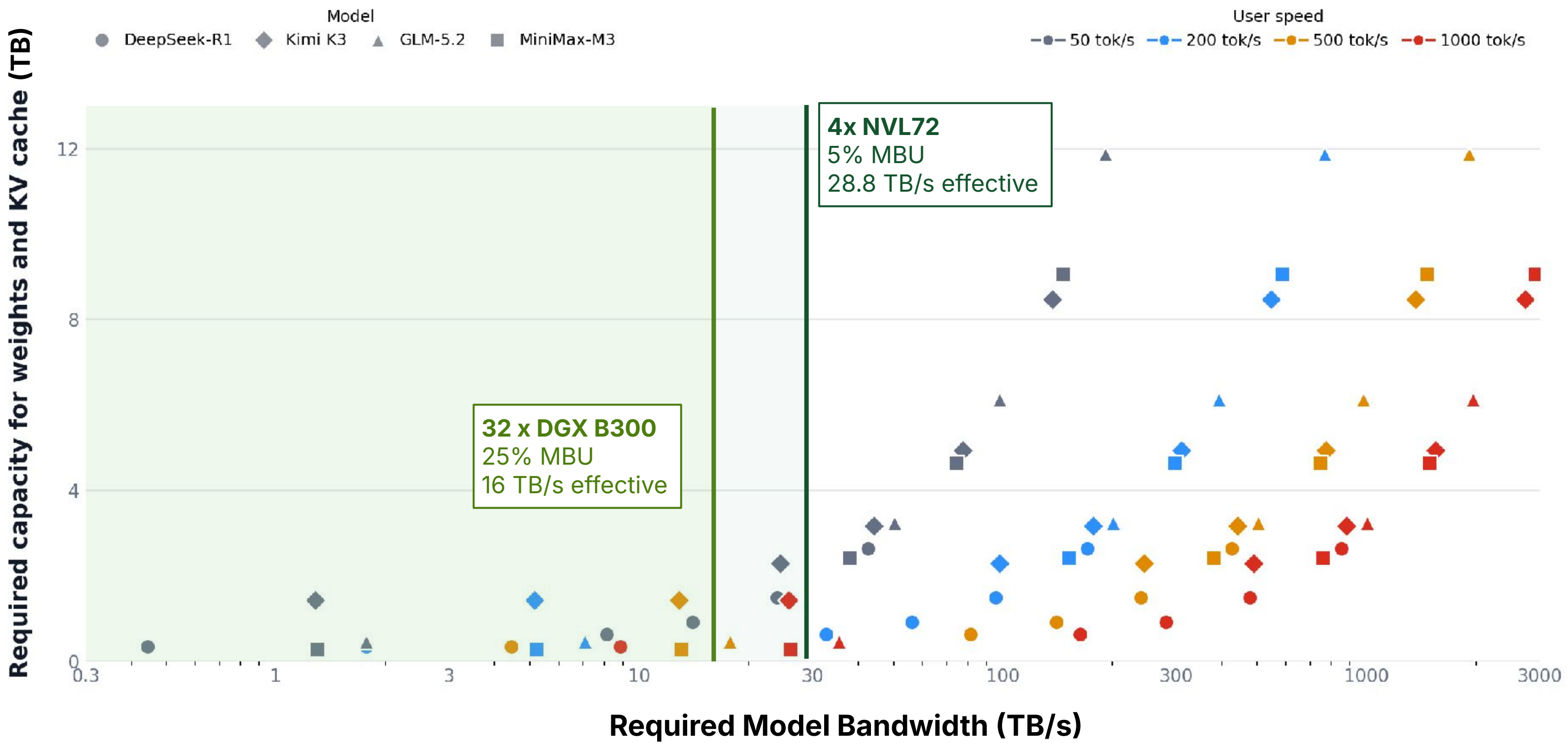
SN50 RDU: High MBU At Scale

DeepSeek-R1 8K/1K | B300 FP4 | 26 InferenceX GPU points | sorted by delivered tok/s/user | RDU inserted by speed



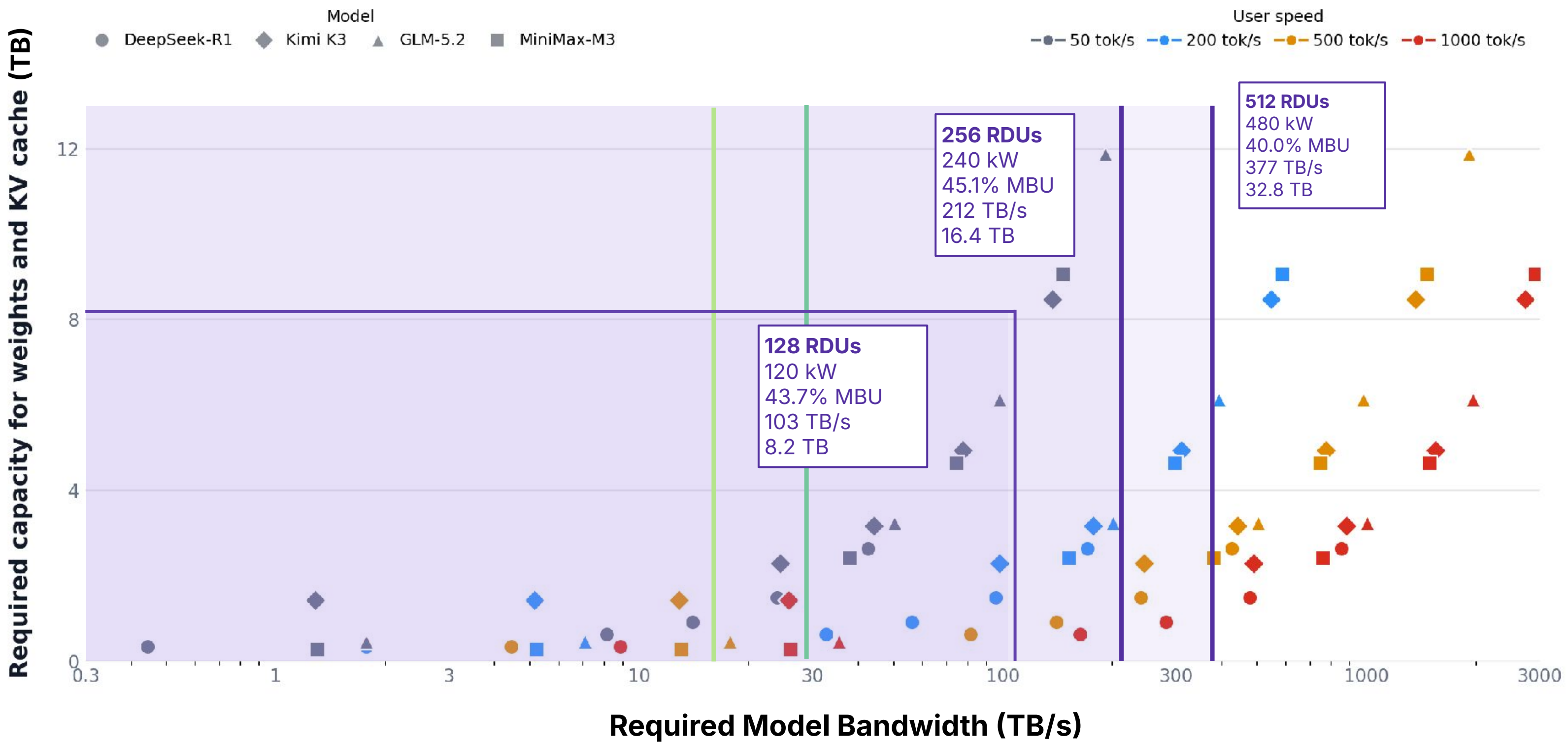
GPUs @500 kW: Power Buys Capacity, not Bandwidth

All frontier-model points | MTP3 accepted length = 3.1

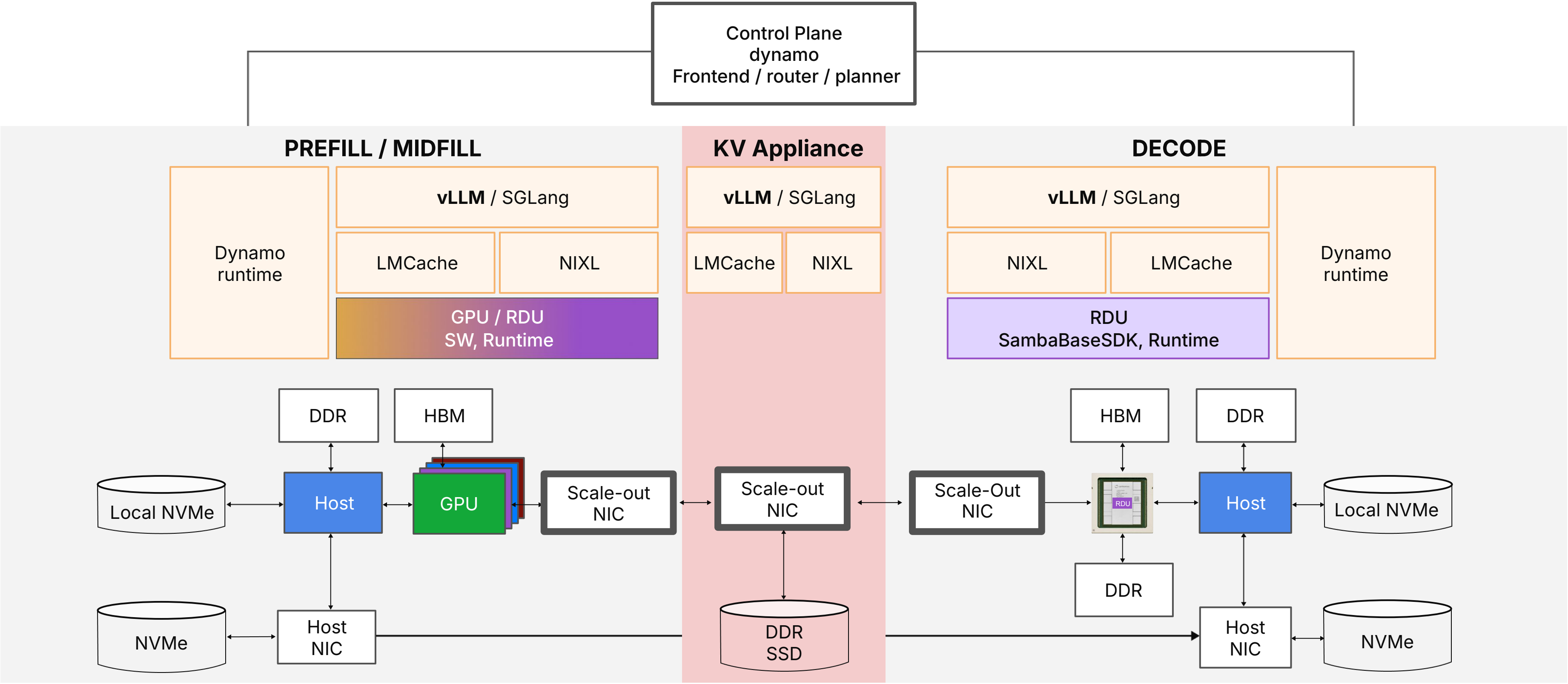


SN50 RDU: Power Scales Capacity and Realizable Bandwidth

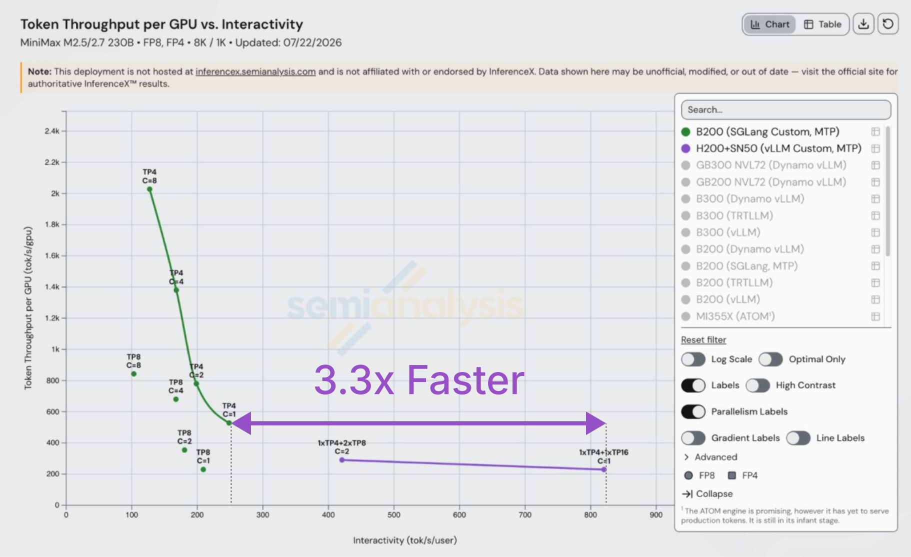
All frontier-model points | MTP3 accepted length = 3.1



Putting it All Together: Heterogeneous Disaggregation



Heterogeneous Disaggregation in Action



SemiAnalysis Benchmarks SambaRack SN50 with Fast Inference on MiniMax M2.7

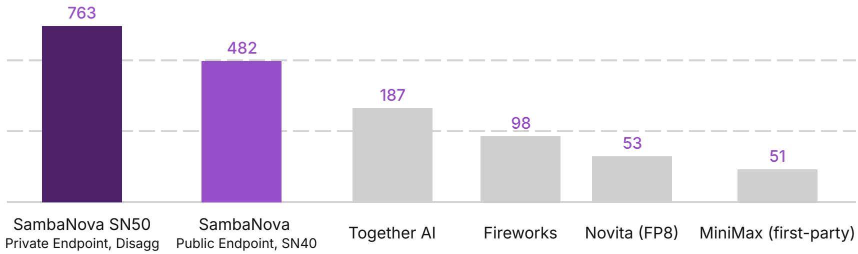
Read More →

Artificial Analysis

INFERENCE BENCHMARK

SambaNova SN50 achieves over 750 tokens/s on MiniMax M2.7

Output tokens per second at 10,000 input tokens, including SambaNova's previous-generation SN40. Higher is better. Comparison figures per Artificial Analysis Serverless API benchmarking (07-07-26).



Deployment: Heterogeneous disaggregated serving

PREFILL **4× NVIDIA H200** → DECODE **16× SambaNova SN50 · TP16**

Engine: vLLM

Output speed · MiniMax-M2.7 · 10,000 input tokens · 32,768 max output tokens per run

Source: Artificial Analysis

AFTER PARTY

The talks are over. The night isn't

The Patio @Rudy's

Tuesday August 25

07:00 PM - 09:00 PM

Palo Alto

Drinks | Apps | Swag | Shuttle from Hot Chips

Luma.com/i6no3a2v